The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

Giulio Ruffini[1,2], Ricardo Salvador[1], Borja Mercadal[1],
Raul de Palma Aristides[2], Francesca Castaldo[1,2],
Nir Grossman[5], Santiago Canals[2,4], and Claudio Mirasso[2,3]

[1]Neuroelectrics, Barcelona, Spain
[2]BCOM, Barcelona, Spain
[3]IFISC (UIB-CSIC), Palma de Mallorca, Spain
[4]Instituto de Neurociencias (UMH-CSIC), Sant Joan d'Alacant, Spain
[5]Department of Brain Sciences and UK Dementia Research Institute,
Imperial College London, London, United Kingdom

E-mail: giulio.ruffini@neuroelectrics.com

Working draft v0.15.0-draft23 2026-10-24

ORCID iDs: G. Ruffini 0000-0003-3084-0177; R. Salvador 0000-0003-1235-6533; B. Mercadal 0000-0002-0941-6796; R. de Palma Aristides 0000-0002-0255-2293; F. Castaldo 0000-0002-4947-4552; N. Grossman 0000-0003-2002-9049; S. Canals 0000-0003-2175-8139; C. Mirasso 0000-0003-2980-7038.

Supplementary Material

Additional methods, derivations, and figures accompanying the article. Sections, equations, and figures use S-prefixed numbers; references to article equations are marked as main-text references.

Notation and units

Table 1 groups the symbols by their role in the calculation. JR and LaNMM voltages are in mV and rates in s^{-1}. QIF voltages and current-like drives are dimensionless model variables; QIF rates are physical rates, integrated in ms^{-1} and converted to Hz where indicated. The uncoupled MPR derivation explicitly uses membrane-time units; the main population equations retain physical time and rate. All frequencies and time constants in a formula must use the same time unit. In the FEM comparison $\mathbf{n}^*$ is the direction maximizing the envelope at balanced currents, held fixed during the ratio sweep. A star otherwise marks a fixed point or, on a bifurcation parameter, its critical value. Subscripts E and I identify excitatory and inhibitory populations; they are not field or current variables.

Table 1: Notation and units. "Normalized" means dimensionless in the stated model scaling; output units depend on the named observable. Equation numbers prefixed by S identify supplementary definitions; B-prefixed numbers identify article-appendix equations, and unprefixed numbers identify main-text equations. Lettered locations identify article appendices.

Symbol	Meaning	Units	Defined in
Field and access			
$E(t), E_0, \mathbf{E}_j$	Applied field, its amplitude, and vector field of tone $j = 1, 2$	V/m	Eq. (3)
$s(t), \varepsilon_j, \varepsilon$	Local polarization; tone amplitudes; common amplitude ($\varepsilon_{\mathrm{TI}}$ and $\varepsilon_{\mathrm{tACS}}$ label the waveform)	mV	Eqs. (1), (4)
$\varepsilon_{\mathrm{D}}, \varepsilon_{\mathrm{A}}, \varepsilon_{\mathrm{eq}}$	Direct DC polarization, signed AC amplitude, and equivalent direct polarization matching a TI source	mV	§2.6
$I_j^{\mathrm{pair}}, \mathcal{P}_j$	Peak electrode-pair current; signed local polarization per unit current	mA; mV/mA	§2.3
$\mathbf{F}_j, \sigma_{\mathrm{cond}}$	FEM field per pair current; tissue conductivity	(V/m)/mA; S/m	§3.3
$\mathcal{A}(t), \overline{\mathcal{A}}$	Nonnegative polarization envelope and its beat-period mean	mV	Eq. (8); Table 1
f_1, f_2, f_c	Tone frequencies and their center	Hz	Eq. (1)
ω_j, ω_c	Angular frequencies $2\pi f_j, 2\pi f_c$	rad/s	Eqs. (1), (4)
$\Delta f, \Omega$	Beat frequency $f_2 - f_1$; $\Omega = 2\pi\Delta f$	Hz; rad/s	Eq. (1)
m	AM-surrogate modulation depth	dimensionless	Eq. (17)
$\Lambda[E], \lambda_0, \lambda_f$	Field-induced polarization; static and frequency-dependent coupling	mV; mV/(V/m)	Eqs. (B1), (4)
$\boldsymbol{\lambda}_0^k, \boldsymbol{\lambda}_f^k$	Coupling vectors for compartment k	mV/(V/m)	Eq. (4)
H, H_k, H_m	Dominant-compartment access transfer, indexed transfer, and point-membrane transfer	dimensionless	Fig. 2; §2.2, §2.8
τ_m, τ	Membrane time; access-filter time in kHz controls	ms	Eqs. (15), (13)
f_{char}	Characteristic frequency $1/(2\pi\tau)$	Hz	Table 2
$\ell(\omega), \ell_0$	Frequency-dependent and static cable space constants	μm	App. S1.1, Eq. (S1)
$\mathbf{V}, \mathsf{C}_m, \mathsf{G}_m$	Compartment voltages, capacitance and conductance matrices	V; F; S	App. S1.1
$\mathbf{I}^0, \mathbf{p}_E$	Endogenous currents; field-to-current vector	A; A/(V/m)	App. S1.1
$\mathbf{x}, \phi_j, \mathbf{n}$	Position, field potential, unit cell orientation	m; V; unit vector	§2.3, §2.5
$\mathcal{I}$	Nonnegative mean-square local polarization before detector gain	mV^2	§2.5
Detection			
$\sigma, \sigma', \sigma''$	JR/LaNMM rate curve, slope and curvature	s^{-1}; $s^{-1}mV^{-1}$; $s^{-1}mV^{-2}$	Eq. (2)

Continued on next page

Table 1 continued

Symbol	Meaning	Units	Defined in
e_0, v_0, ρ	Half the rate ceiling, inflection voltage, inverse voltage scale	s^{-1}; mV; mV^{-1}	Eq. (2)
PEIX	Signed sigmoid saturation index	dimensionless	Eq. (14)
Y_Ω, Y_{DC}	Complex beat amplitude and signed steady source before recurrence	detector-output units	Eqs. (6), (5)
a_j^k, b_j^k, B_{kl}	Compartment polarization, polarization per current, bilinear mixing coefficient	mV; mV/mA; output/mV^2	§2.3
$\chi^{(2)}$	Dynamic coefficient mapping a polarization product to a rate beat	$s^{-1}mV^{-2}$	§2.2
g, T, Y_{LF}	Memoryless response, envelope transduction, carrier-averaged output	specified output units	Eq. (10)
c_1, c_2, c_4	Linear, quadratic and quartic envelope coefficients	output/$(mV)^{1,2,4}$	§2.4
$\mathcal{H}_{det}, \mathcal{H}_{meas}$	Second/fundamental harmonic magnitude ratios before and after the network	dimensionless	Eq. (S3)
$\mathcal{M}[y]$	Peak-to-peak excursion of the quantity y	units of y	Eq. (9)
$F(Q), F', F''$	MPR steady-rate curve and drive derivatives	normalized	Eq. (16)
Q, q	Mean total drive $\bar{\eta} + I$; single-cell drive $\eta + I$	normalized	Eq. (16); App. B.3
I_{ac}	Single-tone current amplitude	normalized	§2.10
κ_{DC}, κ_V	Mean-rate mixing per squared current or squared voltage excursion	normalized	Eqs. (B6), (S2)
d, ν, ω_z	MPR relaxation, ringing and zero-crossing angular frequencies	membrane-time units	App. B.2
$\mathsf{J}, \mathbf{h}, H_v$	Reduced MPR Jacobian, unit-current response and voltage transfer	normalized	§2.10; App. B.3
$\mathbf{b}_2, \mathsf{I}_2$	Quadratic DC source per I_{ac}^2; identity matrix	normalized	App. B.3
$\mathcal{R}_{det}, \mathcal{R}_{lin}$	TI/direct-drive source ratio; linear TI/tACS output ratio	dimensionless	App. S5; §4.7
δ, ζ	Threshold offset; sigmoid rate divided by $2e_0$	mV; dimensionless	§2.4; App. S5
u, ξ, β, k_{ell}^2	Reduced amplitude $\rho\varepsilon$, offset $\rho(v^* - v_0)$, threshold ratio, elliptic parameter	dimensionless	App. S5
$L, \psi, \mathsf{K}, \mathsf{E}$	Unit logistic, normalized waveform, complete elliptic integrals	dimensionless	App. S5
Network			
r_X, v_X	Population firing rate and mean potential, indexed by X	s^{-1}; mV in JR/LaNMM, normalized in QIF	§2.1
t, r, f (MPR derivation)	Time, population rate and single-cell rate in the stated membrane-time convention	normalized locally	App. B.2
$r_{syn}, r_E^{syn}, r_I^{syn}$	Synaptically filtered firing rates	same rate units as input	Eqs. (B5), (19)
$u_{PSP}, u_{X\leftarrow Y}, \hat{L}$	Postsynaptic potential and inverse synaptic operator	mV; maps mV to s^{-1}	Eq. (B1)
A, B, a, b	JR excitatory/inhibitory PSP gains and inverse synaptic times	mV; s^{-1}	§2.1; App. B
h_e, h_i	JR excitatory and inhibitory synaptic impulse responses	mV	Fig. 2
$C, C_{1\ldots4}, C_{X\leftarrow Y}$	JR connectivity scale and pathway weights	dimensionless	§2.1; App. B
y_0, y_1, y_2, v	JR PSPs; voltage readout $v = y_1 - y_2$	mV	App. B
p, p_H, C^*	JR external rate and its upper Hopf value; critical connectivity	s^{-1}; dimensionless for C^*	§3.4, §4.5
$\eta_j, \bar{\eta}, \Delta$	QIF excitabilities, Lorentzian center and half-width	normalized	Eq. (15)

Continued on next page

Table 1 continued

Symbol	Meaning	Units	Defined in
$I, I_E, I_j, A_f, \lambda_I$	QIF input, E-population input, tone amplitudes, forcing amplitude and field coupling	normalized; λ_I in $(\mathrm{V/m})^{-1}$	Eqs. (15), (19)
$\mathcal{U}_j$	Total synaptic and field drive to QIF neuron j	normalized	Fig. 2
J, J^*, w_{XY}	QIF coupling, critical coupling and relative pathway weights	dimensionless	Eq. (19)
$\tau_E, \tau_I, \tau_a, \tau_g, \tau_s$	E/I membrane and excitatory/inhibitory/generic synaptic times	ms	Eqs. (B5), (19)
$\hat{K}_a, \hat{K}_g, H_s$	QIF synaptic rate filters; JR rate-to-voltage transfer	dimensionless; mV s for H_s	§3.5, §2.7
$\mu_\pm, \gamma, \omega_0, f_0$	Modal eigenvalues, damping, angular and ordinary modal frequency	s^{-1} (ms^{-1} where stated); Hz for f_0	Eq. (11)
$\tau_{\mathrm{rel}}, \tau_{\mathrm{rev}}, \mathcal{Q}$	Population relaxation time, $1/\gamma$, and mode quality factor	time; dimensionless for $\mathcal{Q}$	§2.1, §2.7, §5
D, G, R_Ω	Beat-source coefficient, recurrent transfer, complex observed beat response	input/output-dependent	Eq. (12)
θ_j	QIF phase with $V_j = \tan(\theta_j/2)$	rad	§S4.5
$V_j, V_{\mathrm{peak}}, N$	Single-QIF voltage, spike cutoff and neurons per population	normalized; count	§3.6
$\delta\eta_E, \delta\eta_E^*$	Rectified E-drive shift and its Hopf threshold	normalized	§3.6, §4.9
P_1, P_2, r_P, φ_X	Deep/superficial pyramidal populations; JR and LaNMM pyramidal rates	labels; s^{-1} for rates	§4.6; §S6.1
$z, \alpha, \mathcal{F}, \varepsilon_{\mathrm{SL}}$	Stuart–Landau mode, growth rate, forcing and forcing amplitude	normalized mode; s^{-1} for the others	App. S2.4
Measurement			
$L_{\cos}, L_{\sin}, A_\Omega$	Signed lock-in quadratures and nonnegative amplitude	measured voltage or rate units	Eq. (18)
A_{open}	Magnitude of the open-loop rate beat	s^{-1}	§4.5
$w_k, t_k, v_k, \bar{v}$	Hann weights, sample times, voltage samples and weighted mean	1; s; mV; mV	Eq. (18)
$A_{\mathrm{test}}, \theta_0$	Amplitude and phase of the lock-in test tone	readout units; rad	§3.2
$\Delta t, N_{\mathrm{grid}}$	Integration step and number of parameter combinations	s or ms; count	§3.4
$A_\Omega/\langle r_E \rangle$	Beat-locked population modulation depth	dimensionless	§3.6
PPC	Pairwise phase consistency of single-neuron spike phases	dimensionless	§3.6
$f_{\mathrm{out}}, S(f_c)$	Dominant output frequency; LaNMM mean off-alpha lock-in	Hz; mV	App. S7; §S6.1

S1 Carrier access

These calculations support the distinction between applied field and local polarization in §2.8 and its kHz demonstrations.

S1.1 Compartment access and the passive-cable estimate

The passive compartment equation $\mathsf{C}_m\dot{\mathbf{V}} + \mathsf{G}_m\mathbf{V} = \mathbf{I}^0 - \mathbf{p}_E E$ [1] gives the voltage perturbation $\delta\mathbf{V}(\omega) = -(\mathsf{G}_m + i\omega\mathsf{C}_m)^{-1}\mathbf{p}_E E(\omega)$. In SI units, $\mathbf{V}$ contains the compartment membrane potentials (V), C_m and G_m are the capacitance (F) and conductance (S) matrices, $\mathbf{I}^0$ is the endogenous current (A), and $\mathbf{p}_E$ is the field-to-current coupling (A per V/m).

Geometry can also alter the relevant response mode. For a passive cable, the magnitude of the frequency-dependent space constant is

$$\ell(\omega) = \ell_0(1 + \omega^2\tau_m^2)^{-1/4}, \tag{S1}$$

where ℓ_0 is its static value. At 1 kHz and $\tau_m = 16$ ms, this length is about one tenth of its static value, so compartments that are strongly coupled at low frequency can respond differently to the carrier. A somatic charging time alone therefore does not determine every extracellular-field response mode. For a sufficiently long passive cable with a sealed end in a uniform axial field, the end polarization is proportional to this space constant. Each carrier then acquires the factor $(1 + \omega^2\tau_m^2)^{-1/4}$, so the quadratic beat is attenuated by $(1 + \omega_c^2\tau_m^2)^{-1/2}$, about 10^{-2} at 1 kHz and 10^{-3} at 10 kHz for 16 ms. These differ from the point-membrane estimates; soma loading and finite cable length further change the transfer. Thus cable geometry can improve carrier access without changing the somatic membrane time.

S1.2 Measured responses and candidate access sites

Somatic passive time constants are commonly 10–30 ms; reported human pyramidal values include approximately 12 ms [2] and 16.5 ± 3.7 ms [3]. For the illustrative $\tau_m = 16$ ms used here, the power factor $|H_m|^2$ is about 10^{-4} at 1 kHz, 2.5×10^{-5} at 2 kHz, and 10^{-6} at 10 kHz. A static nonlinearity placed after this point-membrane response therefore receives strongly attenuated carriers.

Cohen et al.'s periaxonal circuit provides a reason to distinguish axonal from somatic access [4]. A conducting submyelin space, about 12 nm wide and partly sealed at the paranodes, separates the voltage across myelin from that across the axonal membrane. Their minimum fitted axonal decay was approximately 1 ms. This is a response to injected current in a distributed cable, rather than a measured transfer from a TI field. The AIS is another candidate site, with a high sodium-channel density [5]; this property supports nonlinear sensitivity but does not fix its charging time.

S1.3 Access assumptions in related models

Short access times motivate the model but remain preparation-dependent. The 1 and 0.4 ms cases of [6] precede our effective-time hypothesis. Luff et al.'s pulse-width-modulated TI model uses 1 ms within a 1–100 ms passive-time sweep [7]. The 0.7 μs charging measured by Semenov et al. under strong microsecond field pulses [8] concerns a different response mode from intracellular current charging or periaxonal conduction. A 10–100-fold reduction in QIF membrane time is therefore a sensitivity hypothesis, whose population consequence follows from the slow-synapse limit.

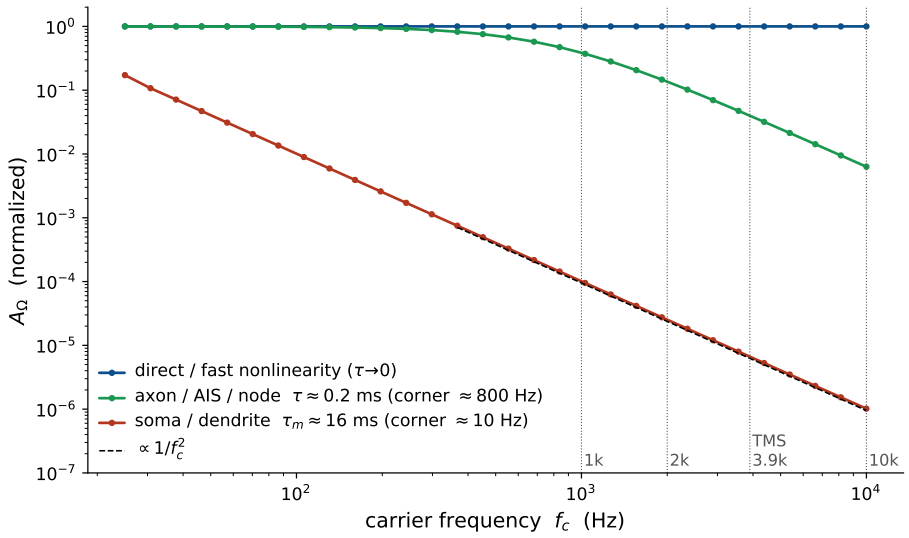


Figure S5: Carrier attenuation depends on the assumed access time (Jansen–Rit, open loop). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide); the assumed fast response ($\tau = 0.2$ ms) retains about 0.14 of the direct response at 2 kHz and 0.04 at 3.9 kHz. These are sensitivity cases, not measurements of anatomical compartments.

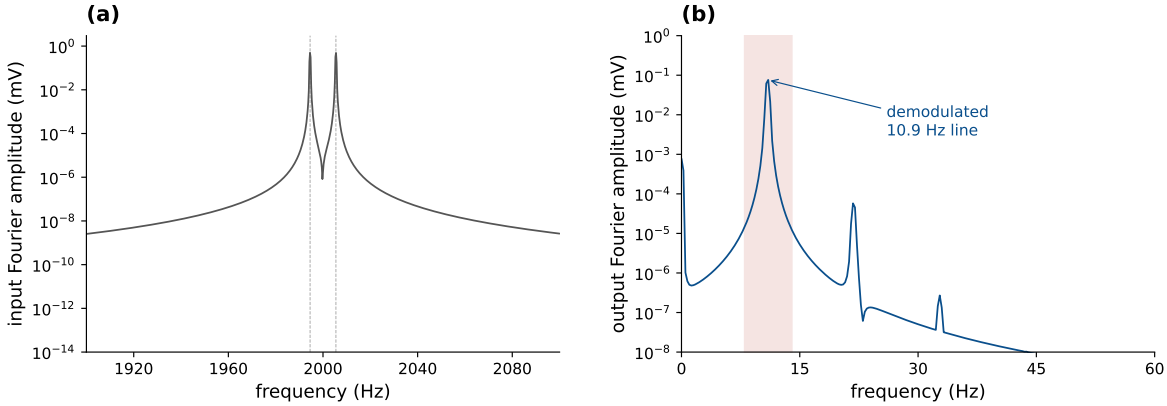


Figure S6: Two-tone TI at a kHz carrier (Jansen–Rit, assumed fast access). Two equal tones at $f_1 = 1994.543$ and $f_2 = 2005.457$ Hz, beating at 10.914 Hz, pass through the assumed low-pass ($\tau = 0.2$ ms) into the recurrent column. *(a)* The input amplitude spectrum has lines only at f_1 and f_2; in the 8–14 Hz band its maximum is 2.2×10^{-13} mV. *(b)* The output carries a demodulated line at the beat, 10.9 Hz.

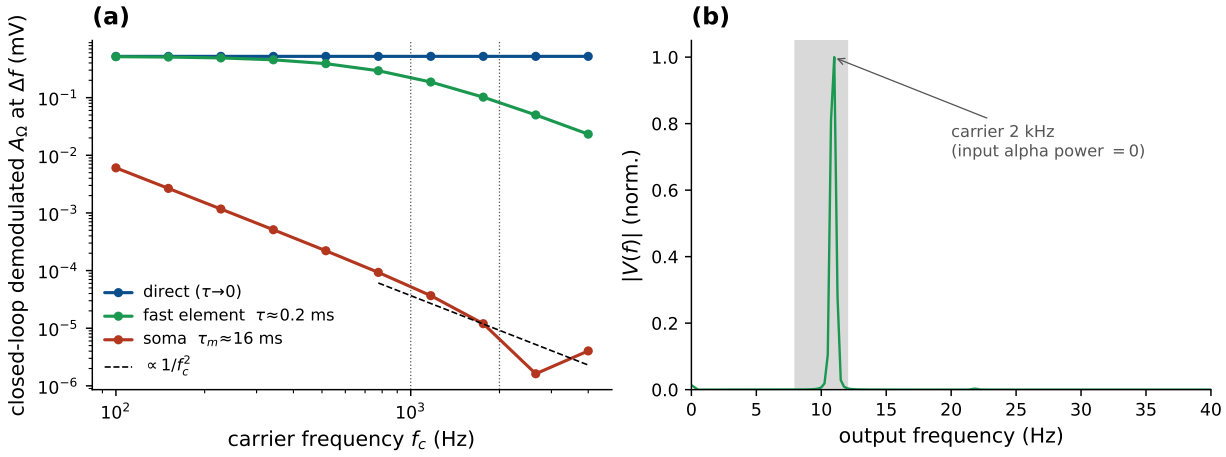


Figure S9: Demodulation at a kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three access models: direct ($\tau \to 0$, carrier-independent), fast ($\tau = 0.2$ ms), and somatic ($\tau_m = 16$ ms; $1/f_c^2$ roll-off). The fast time is assumed rather than assigned to a measured compartment. *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: an alpha line at f_0, synthesized by the column although the input carries no alpha power.

S2 Detection and harmonic signatures

These calculations support the carrier-scaling and harmonic tests in §2.4 and the dynamic detector in §2.10.

S2.1 Fixed-voltage mixing and carrier exponents

Time and frequency use the membrane-time units defined in article Appendix B.2; r^*, d, ν have the same definitions. Membrane access is already included in this calculation. The linear voltage response is $H_v(\omega) = (i\omega + d)/[(i\omega + d)^2 + \nu^2]$. Dividing by its squared magnitude gives the mixing coefficient at fixed voltage excursion,

$$\kappa_V(Q, \omega) = \frac{\kappa_{\mathrm{DC}}(Q, \omega)}{|H_v(\omega)|^2} = \frac{F'(Q)}{2} \frac{\omega^2 + 3d^2 - \nu^2}{\omega^2 + d^2}. \tag{S2}$$

This coefficient tends to $F'/2$ at high frequency. The $1/\omega^2$ suppression at fixed current comes from $|H_v|^2$, the membrane filtering already represented by main-text Eq. (4); it is not an additional detector filter. A static curve placed after the same membrane has the same leading carrier roll-off, although its mixing coefficient and sign can differ. The steady-state curvature is $F'' = 4r^*(3d^2 - \nu^2)/(d^2 + \nu^2)^3$, so the inflection occurs at $\nu^2 = 3d^2$.

For nearby fast tones, the quadratic terms in main-text Eq. (15) contribute at different orders: v^2 at ω^{-2}, rv at ω^{-3}, and r^2 at ω^{-4}. Table S1 compares the carrier and beat routes to the firing rate for a static curve and a dynamic population with the same membrane access.

stage	static curve, same membrane	dynamic population
point-membrane current → voltage	ω^{-1}	ω^{-1}
voltage → rate, carrier	1	ω^{-1}
current → rate, carrier	ω^{-1}	ω^{-2}
two tones → beat in the rate	$(\omega_1\omega_2)^{-1}$	$(\omega_1\omega_2)^{-1}\times$ const.

Table S1: Fast-carrier scaling with matched point-membrane access, at fixed beat frequency and a nonzero leading mixing coefficient. The dynamic population integrates the carrier once more between voltage and rate; the beat is formed before that step. Prefactors depend on the operating point and beat-frequency susceptibility.

The first row gives the shared membrane factor $1/\omega$. The second converts voltage to rate: a static curve adds its slope, whereas the dynamic rate equation integrates the carrier once more. Combining these stages gives the third-row carrier responses, ω^{-1} and ω^{-2}, respectively.

The beat in the last row is formed before that additional rate integration: static curvature and dynamic v^2 each mix the attenuated voltage, retaining $(\omega_1\omega_2)^{-1}$. Equal exponents do not imply equal amplitudes; mixing and the susceptibility at fixed Ω determine the prefactors. For the uncoupled carrier-averaged response, κ_V in Eq. (S2) tends to $F'/2$, but is not the full finite-Ω network prefactor. Thus the dynamic rate attenuates the carrier twice while forming the beat between those stages.

S2.2 Harmonic ladder and mixed detectors

Envelope following and square-law detection are distinguishable at a single site. At a matched target, $\mathcal{A}(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos\theta| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1}\frac{(-1)^{n+1}}{4n^2-1}\cos(2n\theta)$ gives a harmonic ladder whose nonnegative Fourier amplitudes $A_{n\Omega}$ obey $A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}\mathcal{A}^2 = \varepsilon^2(1 + \cos\Omega t)$ contains only DC and Ω. For the mixed family $T(\mathcal{A}) = c_1\mathcal{A} + c_2\mathcal{A}^2$, with $c_1 \neq 0$ and a nonzero fundamental, the pre-network

ratio is

$$\mathcal{H}_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_{\Omega}}\right| = \frac{1/5}{\left|1 + \frac{3\pi}{4}\, c_2\varepsilon/c_1\right|}, \tag{S3}$$

so measuring $\mathcal{H}_{\mathrm{det}}$ constrains $c_2\varepsilon/c_1$ within this family.[1]

S2.3 AM-surrogate and genuine two-tone comparisons

These checks support the waveform choices in §3.1. Agreement at the leading beat does not imply equal steady sources or driven operating states.

Table S2: Waveform comparisons at the stated operating points. JR response ratios use the same voltage readout; timing ratios use the same pyramidal firing rate.

Comparison	Conditions	Result
JR open-loop beat	$v^* = 8$ mV	Two-tone/AM: 1.0000 at 0.02 mV; 0.988 at 1 mV.
JR modulation/mean shift	Pinned sweep: far from Hopf, $\gamma = 1$ s^{-1}, nearest point	Two tones: 14.5, 58, 44; AM: 19, 66, 45. AM has 3/4 of the two-tone direct steady source.
JR TI/tACS output	Far from Hopf, $\gamma = 1.07$ and 0.12 s^{-1}	AM: 0.14, 0.24, 0.47; two tones: 0.14, 0.24, 0.46. Difference at most 2.2% across the grid.
JR matched-source output	Three couplings, settings in §S4.1	TI/tACS: 0.99, 0.94, 0.92 for AM; 0.99, 0.93, 0.90 for two tones.
PING beat	Well below Hopf; $\bar{\eta} = 0.9$	Matched AM and two-tone responses agree within 0.07% and 4%, respectively.

S2.4 The cubic Hopf normal form lacks a quadratic mixer

The Stuart–Landau equation [9]

$$\dot{z} = (\alpha + i\omega_0)z - |z|^2 z + \mathcal{F}(t), \qquad \gamma = -\alpha > 0, \tag{S4}$$

uses seconds for time and a normalized complex mode z, with the cubic coefficient set to 1 s^{-1}. Its linear growth rate is $\alpha = -\gamma$; γ has units s^{-1}. The mode has damping γ and natural angular frequency ω_0, driven by the applied input $\mathcal{F}$ (normalized mode units per second). For a weak input $\mathcal{F} = \varepsilon_{\mathrm{SL}}\psi_{\mathrm{SL}}(t)$ with dimensionless waveform ψ_{SL} on the stable side, the perturbative response has the form $z = \varepsilon_{\mathrm{SL}} z_1 + \varepsilon_{\mathrm{SL}}^3 z_3 + \cdots$. There is no second-order source: the first-order response contains only the applied frequencies, and the cubic first contributes at order $\varepsilon_{\mathrm{SL}}^3$. Thus the normal form has a resonant linear susceptibility but lacks the quadratic difference-frequency generation of main-text Eq. (6). This statement concerns the weak-input expansion; it is not an exclusion of every higher-order intermodulation or strong-drive response.

In the numerical control ($\alpha = -0.5$ s^{-1}, $f_0 = 10$ Hz, $f_c = 100$ Hz, $\varepsilon_{\mathrm{SL}} = 0.5$ s^{-1}), a square-law detector gives a beat-frequency lock-in amplitude of 2.5×10^{-1}, while $\mathrm{Re}\, z$ gives approximately 2×10^{-8}, at the numerical floor. Driving the same mode directly in band recovers the forced $1/\gamma$ growth. The control separates resonant amplification from the nonlinear source required to drive it at the beat.

[1] The inverse has two branches unless the sign of the fundamental relative to the linear term is known; for $c_2/c_1 \geq 0$ it is unique.

S3 Spatial maps

These definitions and checks support the geometric maps in §4.4 and the interpretation of mean-square polarization in §2.5.

S3.1 Envelope-mean subharmonicity

The envelope mean has the same restriction. At each beat phase ϑ, $\mathcal{A}$ is the Euclidean norm of $(\varepsilon_1 + \varepsilon_2 \cos\vartheta,\, \varepsilon_2 \sin\vartheta)$, a vector of harmonic functions. A norm is convex, so this envelope is subharmonic; averaging over ϑ preserves that property. Thus both idealized mean maps lack a strict interior maximum under these conditions (Fig. S13b).

S3.2 Isotropic orientation averages

For isotropically distributed unit orientations $\mathbf{n}$ with orientation-independent access and curvature, $\langle(\mathbf{n}\cdot\mathbf{E}_j)^2\rangle = |\mathbf{E}_j|^2/3$ and $\langle(\mathbf{n}\cdot\mathbf{E}_1)(\mathbf{n}\cdot\mathbf{E}_2)\rangle = \mathbf{E}_1\cdot\mathbf{E}_2/3$: orthogonal fields cancel the averaged beat, but retain the DC intensity.

S3.3 Head-model procedure

To compare the spatial predictions on the same anatomy, we compute all maps in one open-access subject, IXI002 of the IXI dataset[2] (T1-weighted MRI; female, 35 years). The T1 image was segmented into six tissues (scalp, skull, cerebrospinal fluid, gray matter, white matter and eyes) and meshed with SimNIBS 3.2.3 `headreco` [10], with its CAT12 option [11]. Tissue conductivities were 0.33 (scalp), 0.008 (skull), 1.79 (cerebrospinal fluid), 0.40 (gray matter), 0.15 (white matter) and 0.33 S/m (eyes) [12]. These are low-frequency values; no dispersion correction was applied at the kHz carriers. The volume conductor is linear and quasi-static, so the carrier frequencies do not enter the maps.

The electrodes follow the published hippocampal montage of Violante et al. [13], placed by its recipe on the subject's registered 10–10 positions. Pair 1 (anterior) joins e1, 25 mm from T9 toward FT9, to e2, 10 mm from AF8 toward Fp2. Pair 2 (posterior) joins e3, 25 mm from T9 toward TP9, to e4, 10 mm from P8 toward PO8. Each electrode is a 20-mm disc of 3 mm of gel (4.0 S/m) under 2 mm of metal (10 S/m), meshed into the head model with SimNIBS 3.2.3; the resulting mesh has 3.64×10^6 linear tetrahedra. As a montage sensitivity, the 10–10 rendering of this montage, FT7–F8 (anterior) and TP7–P8 (posterior), which Yatsuda et al. [14] used as their starting montage, was meshed and solved in the same way. Each montage has its own mesh, carrying only its own four electrodes.

Each pair was solved once at 1 A. The potential ϕ_j (V) satisfies $\nabla\cdot(\sigma_{\text{cond}}\nabla\phi_j) = 0$, where σ_{cond} is tissue conductivity (S/m). Current enters and leaves through each electrode's metal contact as a Neumann condition, distributed in proportion to nodal area; the rest of the scalp carries no normal current, and one pinned node fixes the gauge. The linear finite-element system was assembled and solved with Neuroelectrics' temporal-interference library, by conjugate gradients with algebraic-multigrid preconditioning, to a relative residual of 10^{-8}. The field $-\nabla\phi_j$ was recovered at gray- and white-matter nodes by superconvergent patch recovery [15], separately within each compartment (446,877 node–compartment pairs). Dividing these fields by the solve current gives the field-per-current vectors $\mathbf{F}_j$, expressed in (V/m)/mA to match I_j^{pair}. Every current condition is a scaling of these two fields. With peak pair currents I_1^{pair} (anterior) and I_2^{pair} (posterior) summing to 4 mA, the ratio 1:1 is 2 + 2 mA and 1:3 is 1 mA anterior plus 3 mA posterior, the two conditions of Violante et al. The ratio sweep sets the posterior share $I_2^{\text{pair}}/(I_1^{\text{pair}} + I_2^{\text{pair}})$ from 0.1 to 0.9 in steps of 0.025.

[2] `https://brain-development.org/ixi-dataset/`

The field-to-polarization conversion follows §2.3: $\varepsilon_j(\mathbf{x}) = I_j^{\text{pair}}\mathcal{P}_j(\mathbf{x})$, with $\mathcal{P}_j = \lambda_0\,\mathbf{F}_j\cdot\mathbf{n}$, one uniform static coupling $\lambda_0 = 0.5$ mV per V/m [16], and $\lambda_f \approx \lambda_0$. A uniform coupling multiplies the envelope map by λ_0 and the two second-order maps by λ_0^2. Target-to-reference ratios, normalized maps and focus positions do not depend on it. Hippocampal couplings of 0.12–0.18 mV per V/m [17, 18] would scale the absolute values by 0.24–0.36 and 0.06–0.13, respectively.

The detector compartment fixes the projection direction. A laminar direction for a pyramidal axon initial segment, a local axon direction for a node, or an orientation distribution for terminals defines a different access model. Isotropic averaging is one possible distribution, not an anatomical consequence of choosing terminals. Maximizing a geometric map over orientation gives an upper bound for that map at each location; it does not identify the detector's orientation.

The projection direction $\mathbf{n}(\mathbf{x}) = \mathbf{n}^*(\mathbf{x})$ is the envelope-maximizing direction at the reference ratio 1:1: the closed-form maximizer, over unit vectors, of $2\min(|\mathbf{F}_1\cdot\mathbf{n}|, |\mathbf{F}_2\cdot\mathbf{n}|)$ [19]. It is chosen once at each node and then held fixed for 1:1, 1:3 and every ratio of the sweep, and all three maps use it. Along it, the 1:1 envelope map equals its maximum over orientation. At any other ratio it is the envelope seen by an element with that fixed orientation, which can be below that ratio's own maximum. As an orientation sensitivity, every statistic was repeated along a second fixed field, the radial direction from the brain centroid. The isotropic comparison averages the signed mixing term over uniformly distributed orientations before its magnitude is taken: $\langle\varepsilon_1\varepsilon_2\rangle_{\mathbf{n}} = \lambda_0^2 I_1^{\text{pair}} I_2^{\text{pair}}\,\mathbf{F}_1\cdot\mathbf{F}_2/3$, so the isotropic beat excursion is $\frac{2}{3}\lambda_0^2 I_1^{\text{pair}} I_2^{\text{pair}}|\mathbf{F}_1\cdot\mathbf{F}_2|$. Elements whose beats have opposite phase cancel in this average.

The target is the left hippocampus of a FreeSurfer 6.0.0 `recon-all` segmentation of the same T1 image (`aseg` label 17, 3903 mm^3) [20]. The region of interest (ROI) is the 1,568 gray- and white-matter nodes inside it. The reference is the 273,544 gray-matter nodes outside the ROI, including other deep gray matter. It is a gray-matter reference, not a cortex-only contrast. The mesh and the segmentation share one image frame; tissue-volume, centroid and white-matter checks confirmed their alignment. As a label sensitivity, the Neuromorphometrics atlas hippocampus, warped into the subject with SimNIBS 3.2.3 `mni2subject` and the subject's nonlinear MNI transform (1,345 nodes; Dice 0.72 with the FreeSurfer label), replaced the ROI. As a tissue sensitivity, the ROI was restricted to its gray-matter nodes (1,174 with the FreeSurfer label, 1,013 with the atlas label).

The target-to-reference ratio is the ROI 95th percentile divided by the reference 95th percentile; its mean-based companion divides the ROI mean by the same reference percentile. All percentiles and centroids weight nodes equally. The quadratic excursion and $\mathcal{I}$ are second order in the field, whereas the envelope excursion is first order. To separate this difference in degree from the spatial weighting, each second-order statistic is also given degree-matched, computed on the square root of the map. This is a geometric comparison; the raw maps remain the respective detector predictions before gain. The focus is the centroid of the top 10% of ROI nodes, ranked by the map's own value, with 5, 25 and 50% as a sensitivity. Ranking makes the focus exactly invariant to a uniform positive factor. Its displacement from 1:1 is measured along the hippocampal long axis, the first principal axis of the ROI nodes, with anterior positive. Field projections still use $\mathbf{n}^*$, independently of this reporting axis. The 1:1 to 1:3 displacement was computed for every combination of montage, label, ROI tissue definition and focus fraction (32 per map), each montage along its own 1:1 direction.

No signed DC-source or mean-response map is computed. For a spatially uniform detector coefficient, $Y_{\text{DC}} = \frac{1}{2}\sigma''(v^*)\mathcal{I}$ (§2.5) is $\mathcal{I}$ rescaled, with the sign of $\sigma''(v^*)$. A nonuniform coefficient would require an operating-point map, which a head model does not provide, and a mean response would additionally require a circuit model. In particular, $\sigma''(v^*)$ is curvature at the population model's operating point, not a measurement of the transfer at the axon initial segment. Assigning that coefficient to an anatomical detector requires a calibrated access and input–output model.

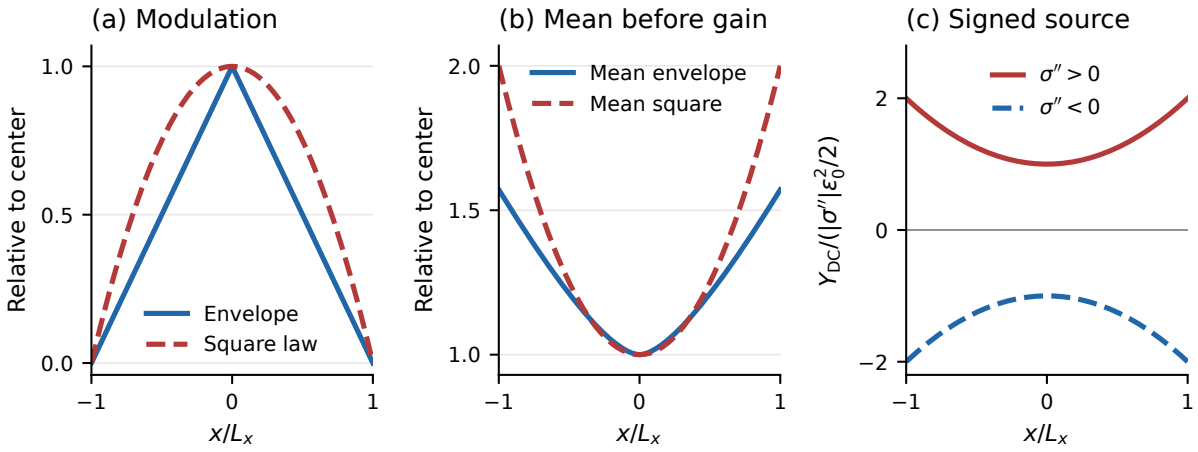


Figure S13: Envelope and quadratic spatial maps. Illustrative line profile: $\varepsilon_1(x) = \varepsilon_0(1 + x/L_x)$ and $\varepsilon_2(x) = \varepsilon_0(1 - x/L_x)$ on $-L_x \le x \le L_x$, with polarization scale $\varepsilon_0 > 0$ (mV) and length $L_x > 0$ (m). In (a,b), each geometric curve is divided by its own positive value at $x = 0$. *(a)* Modulation is $1 - |x/L_x|$ for envelope following and $1 - (x/L_x)^2$ for quadratic detection: both peak centrally, but the quadratic profile is broader here. *(b)* The mean envelope and mean-square polarization $\mathcal{I}$ both increase toward the edges; neither includes a signed detector gain. *(c)* The static JR rate source $Y_{\mathrm{DC}} = \sigma''\mathcal{I}/2$ for uniform positive curvature (solid red) or negative curvature of the same magnitude (dashed blue). Both use the same positive scale $|\sigma''|\varepsilon_0^2/2$, giving $\pm[1 + (x/L_x)^2]$; the gray line marks zero. These are assumed detector states before recurrent feedback, not inferred physiological responses. This is a line profile, not a head model or a demonstration of three-dimensional focusing.

S3.4 Sensitivity of focus displacement

The spatial ordering in the article persists across montage, labeling, tissue, and focus-fraction choices. Table S3 gives the ranges; orientation changes absolute contrast more than the atlas label in this subject. For the reference projection and label, the top-5–50% focus moves 2.2–4.0 mm anteriorly for the envelope and 2.4–8.2 mm posteriorly for mean-square polarization.

Table S3: Focus displacement from 1:1 to 1:3. Envelope shifts are anterior; mean-square shifts are posterior. The quadratic beat shift is zero in every case. Contrasts quoted in the last column are raw 1:1 envelope/quadratic contrasts.

Sensitivity	Envelope (mm)	Mean square (mm)	Contrasts
Atlas label	2.0–3.8	3.2–5.9	1.17 / 1.04
Radial direction	1.9–2.8	2.0–3.7	1.62 / 1.97
All 32 combinations	0.7–8.1	0.5–10.6	

S3.5 Mean-square change under current-ratio steering

The mean-square polarization changes by $\mathcal{I}_{1:3}/\mathcal{I}_{1:1} = (\mathcal{P}_1^2 + 9\mathcal{P}_2^2)/[4(\mathcal{P}_1^2 + \mathcal{P}_2^2)]$, exceeding one only where $|\mathcal{P}_2| > \sqrt{3/5}|\mathcal{P}_1| \approx 0.775|\mathcal{P}_1|$. Its hippocampal mean rises by 24% and its focus moves 4.5 mm posteriorly. Equal summed currents therefore do not match local RMS polarization.

S4 Recurrent amplification and numerical checks

These protocols support the coupling, timing, and field-induced gamma results in §4.5–§4.9.

S4.1 JR coupling-sweep settings

The coupling sweeps of Figs. 10 and S11, and the transcranial alternating-current stimulation (tACS) control of Fig. S3, use the single JR column with the field-free operating point v^* pinned: for each global connectivity C we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds v^* (hence $\sigma''(v^*)$) fixed, vary C toward the alpha Hopf, and read γ, ω_0 from the 6×6 Jacobian. The field is the AM surrogate of main-text Eq. (17) for Fig. 10, genuine two tones for the timing comparison (Fig. S11), or a direct in-band sinusoid $s(t) = \varepsilon \cos \Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S3), each read out by the lock-in of v at Δf, except in the timing comparison, which reads the modulation and mean of the same pyramidal rate $r_P = \sigma(v+s)$. That comparison uses 18 s settling and a 20 s window for each waveform. The Figure 10–Figure S3 response-ratio comparison uses $v^* = 8.30$ mV, TI surrogate amplitude 0.3 mV with $m = 1$ and $f_c = 100$ Hz, and tACS amplitude 0.1 mV, driven at each field-free modal frequency. The original TI amplitudes are interpolated logarithmically in C onto the tACS grid. For the waveform comparison, both the surrogate and two-tone TI are rerun on that grid with 12 s settling and a 6 s window. Pinning fixes the background curvature; it does not cancel the stimulation-induced mean shift. Additional controls at $C = 110$, 133.9091, and 136.3 use a tACS amplitude of 0.0143039 mV, matched to the leading TI beat source. They record mean voltage, mean rate, and lock-in with 12 s settling and a 6 s window, repeated with 60 s settling and a 20 s window; the latter settings also test genuine two tones.

S4.2 PING coupling-sweep settings

We compute the linear reference's second-order source from the PING quadratic terms at the central carrier, 300 Hz, and apply the full linearized network response over 15–90 Hz in 0.025 Hz steps. Using the distinct AM sidebands changes the peak gain by less than 0.1% across this sweep.

The simulated coupling sweep uses the AM surrogate, $A_f = 8$, with 5 s settling and a 4 s window. We search around each linear resonance, extending the frequency grid until each sampled maximum is bracketed, and compare amplitudes $A_f = 2, 4, 8$ at nine couplings. Local parabolic interpolation changes the sampled peaks by less than 1.3%. The figure scales the linear reference to the median measured gain over $14.5 < J < 16$; the amplitude check instead normalizes peak response divided by A_f^2 and linear gain to its value at $A_f = 2$, $J = 15.53$. These normalizations compare gain variation with the detector coefficient allowed to vary along this unpinned sweep.

S4.3 Stability of the driven PING endpoint

We check the stability of the driven J-curve endpoint by integrating the full variational matrix along its trajectory. At carrier 300 Hz and beat 50.5 Hz, the AM input repeats every 2 s. After 8 s of settling, we compute the perturbation multipliers over that period and repeat over the next period. Moduli below one indicate decay of perturbations to the periodic response. Two-tone controls at $J = 16.5663$ and 16.5901, both with a 50.5 Hz beat, use 5 s settling, direct perturbation decay, and an 8 s finite-time Lyapunov estimate.

Although a DC-only averaged calculation predicts an unstable endpoint, the full periodically forced solution remains attracting: its largest perturbation multiplier is 0.034 over the 2 s forcing period. Increasing settling from 0.6 to 3 s changes the endpoint lock-in by 0.8%. With genuine two tones, the 50.5 Hz endpoint control also attracts nearby trajectories: perturbations decay at -0.0014 ms^{-1}, and the 8 s Lyapunov estimate is -0.00079 ms^{-1}. These controls test a nonlinear driven response, not linear gain about the DC-shifted equilibrium.

S4.4 Finite-network numerical checks

At $A_f = 10$ and $\bar{\eta} = 0.85$, the finite-network beat depth is 0.093 for the published cutoff, versus the mean-field value 0.128. Raising the spike cutoff from $V_{\text{peak}} = 100$ to 200 and 400 gives depths 0.127 and 0.139, while the mean-rate increase stays at 2.8–2.9%. At the original cutoff, doubling the population gives 0.083 and halving the time step gives 0.093. The response is therefore sensitive to the finite spike cutoff. Halving the step at cutoffs 200 and 400 gives depths 0.121 and 0.141, respectively: cutoff dependence persists, and quantitative convergence remains unestablished at this population size. At the plotted cutoff, the pooled spike-phase vector strength is 0.046; median per-neuron phase consistency remains negative in field-off and field-on windows at all three cutoffs. In the $A_f = 35$ convergence checks, halving the time step changes the lock-in and the pooled vector strength by under 1%; doubling the spike-detection threshold changes the lock-in by about 1% and the mean rate by 2%, but the pooled vector strength at $\bar{\eta} = 0.85$ by about 45%, so phase concentration, unlike rate modulation, is sensitive to how spike times are defined.

S4.5 A cutoff-free theta-neuron check

The change of variables $V_j = \tan(\theta_j/2)$ converts the QIF equation to

$$\tau_X \dot{\theta}_j = (1 - \cos\theta_j) + (1 + \cos\theta_j)(\eta_j + U_X(t)),$$

where U_X is the same recurrent and external input as in the voltage model. An upward crossing of $\theta = \pi$ emits a spike; wrapping by 2π replaces the finite voltage cutoff and reset. We linearly interpolate the crossing time for phase statistics. Empirical spike counts still drive the original second-order synapses. Excitability quantiles, initial voltage, coupling, carrier, and readouts are unchanged.

The pilot used $\bar{\eta} = 0.85$, $A_f = 10$, $f_c = 300$ Hz, a 55 Hz beat, and 800 ms settling. Before running it, we set a stopping rule: rate depth must change by less than 5% under separate step halving, population doubling, and window doubling before extending this solver to carrier controls. Table S4 gives every tested setting. Step and window changes satisfy that criterion; population doubling does not, so the pilot stops. No finite-network equal-RMS control was run.

Condition	N per population	Step (ms)	Window (ms)	Mean E rate (Hz)	Beat depth
Field off	8000	0.0025	800	50.4	0.006
Two tones	8000	0.0025	800	51.8	0.156
Half step	8000	0.00125	800	51.8	0.155
Double population	16000	0.0025	800	51.8	0.137
Double window	8000	0.0025	1600	51.8	0.151
Exact mean field	∞	0.0025	800	52.1	0.128

Table S4: Cutoff-free weak-drive checks. Depth is the rate lock-in divided by mean rate. Relative to the two-tone baseline it changes by −0.4%, −11.9%, and −3.0% under step, population, and window changes. Only the baseline settings have a matched field-off run.

Removing the cutoff raises the baseline depth from 0.093 to 0.156. The three cutoff results are compatible with an inverse-cutoff correction: a fit of $d_\infty - c/V_{\text{peak}}$ gives $d_\infty = 0.156$, with residuals below 0.002. This agreement supports a cutoff contribution but does not establish joint numerical convergence. Doubling the population moves the theta result toward the exact mean field, while also increasing the largest sampled Euler step in phase from about 1.5 to 3.2 radians. A combined population and time-step limit remains untested.

Pooled spike-phase vector strength rises from 0.002 without stimulation to 0.068–0.077 in the driven runs. Median per-neuron phase consistency remains negative, from −0.015 to −0.005, and supplies no evidence for strong individual phase locking. The finite network therefore supports the

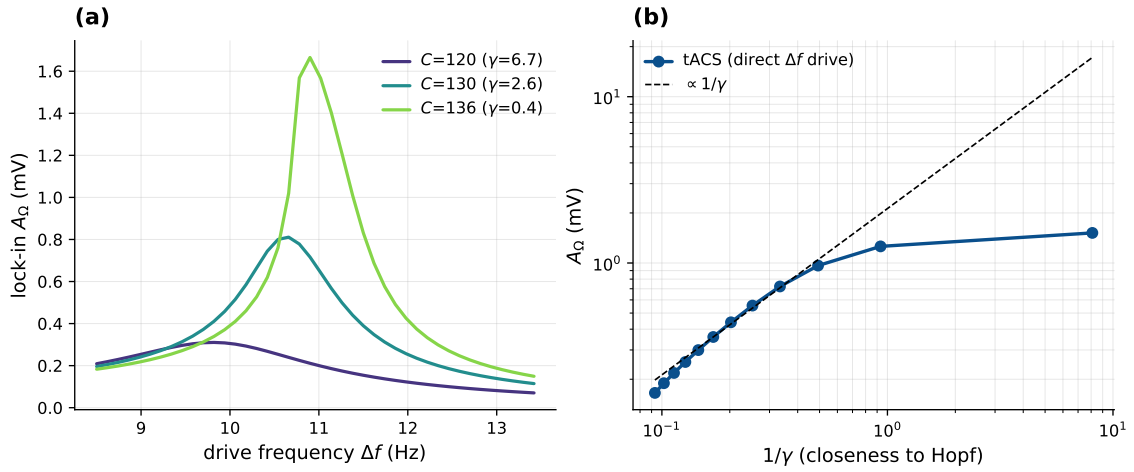


Figure S3: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the field-free operating point pinned, sweeping the coupling C toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 10). The leading tACS response is linear in the field; the TI beat is quadratic. Their response ratio is independent of damping in linear response and changes at the finite amplitudes nearest the Hopf (§4.5). Here $\gamma > 0$ describes the field-free state.

presence of population-rate modulation; its quantitative weak-input depth and agreement with the mean field remain unresolved.

S4.6 Fast-carrier estimate at the illustrated amplitudes

The $A_f = 35$ runs are finite-amplitude illustrations: the fast-carrier estimate gives a voltage excursion $A_f/(\tau_E\omega_c) \approx 1.2$ per tone in the model's voltage units. Its rectified excitatory-drive shift is about 1.5, compared with 0.12 at $A_f = 10$. At $\bar{\eta} = 0.85$, the Hopf threshold for a drive shift applied only to the excitatory population is $\delta\eta_E^* \approx 0.286$. The larger field is therefore expected to cross this threshold, whereas the smaller one remains below it.

For two nearby carriers, the fast-carrier approximation gives a constant shift of the excitatory drive, $\delta\eta_E \simeq A_f^2/(\omega_c\tau_E)^2$, from the quadratic voltage term. We refine this estimate using the full PING linear carrier responses and their second-order DC source, then solve for the carrier-averaged equilibrium and its stability, including the source's dependence on that equilibrium. This approximation isolates the mean shift; direct two-tone integrations retain the beat forcing and provide a separate check.

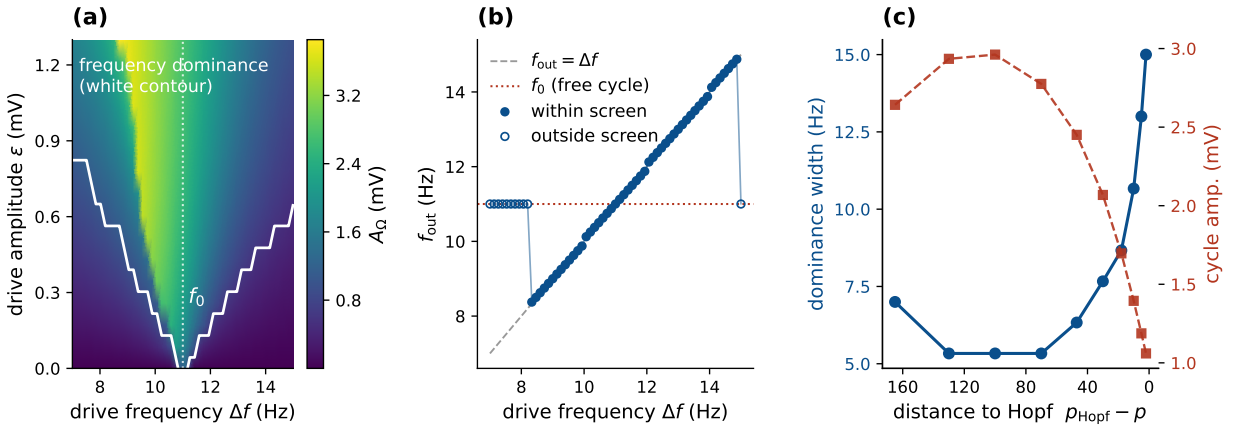


Figure S4: Frequency dominance in the driven autonomous Jansen–Rit column. The column is set in its autonomous alpha limit cycle ($p = 250$, $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid. *(a)* The set $|f_{out} - \Delta f| < 0.3$ Hz forms a wedge near f_0 that widens with drive. It screens for phase capture but does not establish it. *(b)* Inside this region the dominant frequency follows Δf; outside, it remains near f_0. *(c)* Toward the Hopf at fixed drive the dominance width grows as the autonomous cycle amplitude vanishes (blue, left axis; red dashed, right axis). Above the Hopf this width describes frequency selection in a driven autonomous oscillator; it is distinct from the linear susceptibility of the stable-focus side (Fig. 10). Both diagnostics increase toward criticality in these sweeps.

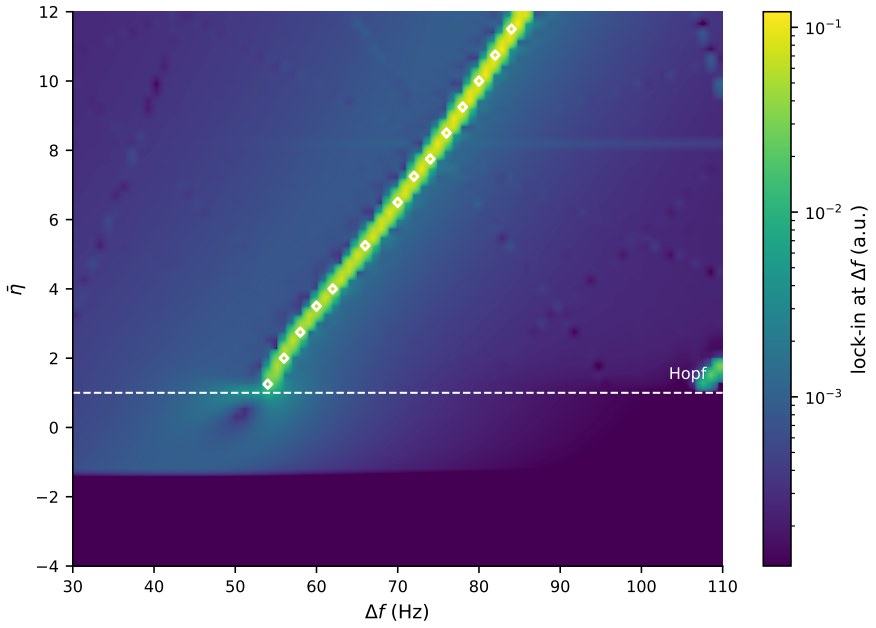


Figure S8: NMM2 PING resonance map (exact mean field). Demodulated response over the (Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency, with f_0 rising with $\bar{\eta}$ for these parameters. White markers outline the frequency-dominance set (§S7), drawn only above the Hopf (dashed line) where there is an autonomous rhythm to capture; below it the same color field is a forced response. The markers do not establish phase capture. Lock-in amplitude cannot separate driven and autonomous contributions, since both can place power at Δf.

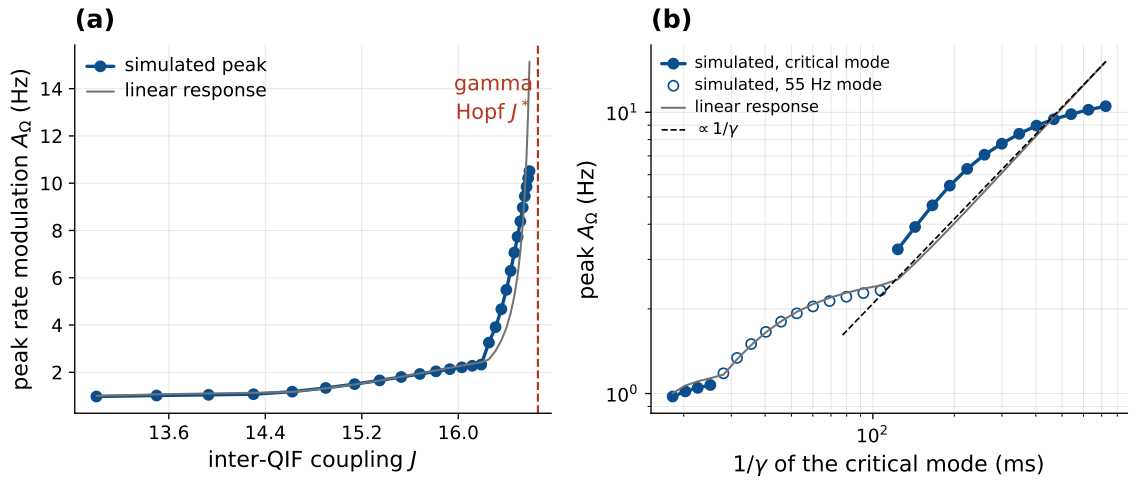


Figure S10: Coupling changes the dominant mode and its gain in NMM2. Unpinned sweep at $\bar{\eta} = 0$, AM surrogate $A_f = 8$. *(a)* Simulated rate peak (blue) and linear reference including the second-order source (gray), normalized over $14.5 < J < 16$. Red dashed: field-free Hopf. *(b)* Gain against inverse damping of the critical mode. Filled symbols: that mode carries the linear peak; open symbols: the competing 55 Hz mode carries it ($14.6 \lesssim J \lesssim 16.2$). The dashed $1/\gamma$ guide is anchored at the last linear point. The simulation exceeds then falls below the guide. The driven endpoint remains attracting under both AM and genuine two-tone forcing.

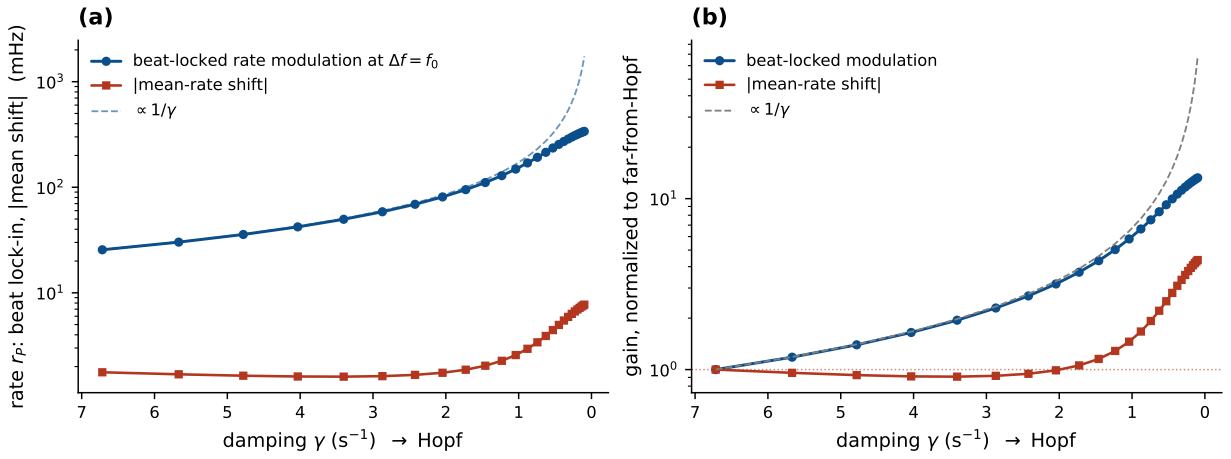


Figure S11: Beat modulation grows more than the mean-rate shift. JR coupling sweep with fixed field-free curvature and genuine two-tone input, $\Delta f = f_0$. Both observables use the pyramidal rate $r_P = \sigma(v + s)$, in mHz. *(a)* Beat amplitude (blue) follows $1/\gamma$ (dashed) before saturating; mean-shift magnitude (red) remains smaller. *(b)* Normalization to the far-from-Hopf values gives growth factors 13.2 and 4.4. Their ratio rises from 14.5 to 58 near $\gamma = 1$ s^{-1}, then falls to 44 as the amplified response rectifies. The AM surrogate gives ratios 19, 66, and 45.

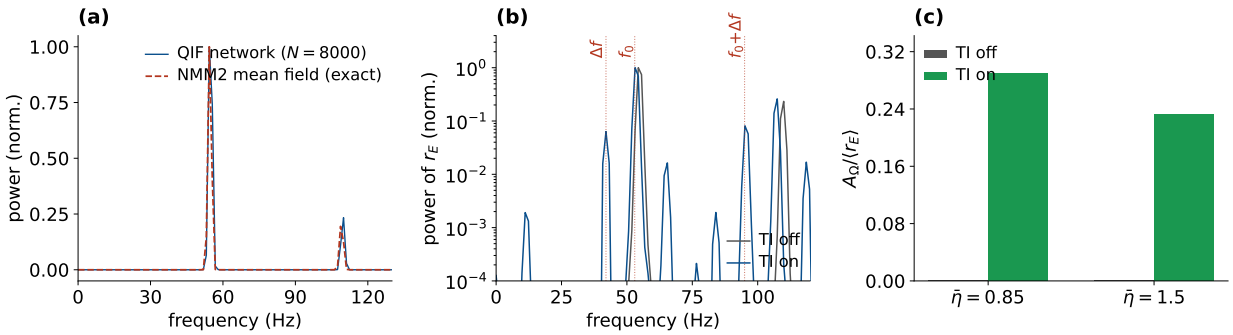


Figure S12: Beat-locked modulation and mean-rate change in the QIF network. *(a)* Field-free power spectrum of $r_E(t)$, normalized: the spiking network ($N = 8000$) matches the exact NMM2 mean field in gamma frequency (∼55 Hz); a normalized spectrum does not test the mean rate. *(b)* Intermodulation above the Hopf (TI off vs. on, log power): the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the persisting gamma near 53–54 Hz, raising the direct Δf line and the sum sideband near 95 Hz, both absent with TI off. These sidebands establish nonlinear mixing; they do not uniquely identify gamma-amplitude gating. *(c)* The Δf modulation depth $A_\Omega/\langle r_E \rangle$, off vs. on: below 0.003 rising to 0.29 ($\bar{\eta} = 0.85$) and 0.23 ($\bar{\eta} = 1.5$), while the mean rate rises by about 8–11%. The labels identify field-free states; both field-on responses contain gamma activity beyond the beat component. Spectra use the same post-settling windows as the rate measurements and are normalized to each spectrum's maximum. Modulation depth is reported rather than the off→on lock-in ratio, whose denominator is the field-free floor.

S5 Efficiency of a memoryless transfer against a direct in-band drive

For the logistic family of memoryless transfers, compare balanced TI tones of polarization amplitude ε each with a direct in-band drive of amplitude 2ε, matching peak polarization. Their leading drive amplitudes are $|\sigma''|\varepsilon^2/2$ and $2\sigma'\varepsilon$, so $\mathcal{R}_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$. The weak-input formula concerns the logistic family. The step-mixture argument below extends the exact carrier-averaged bound to bounded nondecreasing memoryless transfers, with the same operating point, matched peak polarization, and open-loop readout. It does not cover the dynamic detector of §2.10.

Writing $\zeta = \sigma/(2e_0) \in (0,1)$ gives $\sigma' = 2e_0\rho\zeta(1-\zeta)$ and $\sigma'' = 2e_0\rho^2\zeta(1-\zeta)(1-2\zeta)$. Since $|\mathrm{PEIX}| = 1 - 4\zeta(1-\zeta) = (1-2\zeta)^2$, the leading ratio satisfies

$$\mathcal{R}_{\mathrm{det}} \simeq \frac{\rho\,\varepsilon}{4}\sqrt{|\mathrm{PEIX}|} \;\le\; \frac{\rho\,\varepsilon}{4}. \tag{S5}$$

It vanishes at the inflection, where the quadratic coefficient is zero, and approaches its bound in the saturated tails. This maximizes a ratio rather than an absolute response: both drives vanish in those tails. For the JR slope $\rho = 0.56\ \mathrm{mV}^{-1}$, the perturbative bound is 1.4% at $\varepsilon = 0.1$ mV and 0.28% at $\varepsilon = 0.02$ mV.

Optimizing over the sigmoid slope gives a different limit from the weak-input bound. The logistic transfer depends on voltage only through $\rho(v - v_0)$, so writing

$$u = \rho\,\varepsilon, \qquad \xi = \rho\,(v^* - v_0), \tag{S6}$$

we have $\sigma(v^* + \varepsilon\psi(t)) = 2e_0\,L(\xi + u\,\psi(t))$ for any normalized waveform ψ, with $L(\xi) = (1+e^{-\xi})^{-1}$. Every Fourier amplitude, and hence their ratio, is a function of (u,ξ) alone; the factor $2e_0$ cancels. Maximizing over v^* and ρ at fixed ε is therefore the same as maximizing over (u,ξ), and the family supremum is independent of field strength. Evaluating the exact cycle response rather than its quadratic truncation and optimizing ξ at each u, the ratio rises monotonically over the range explored and converges to the hard-threshold limit

$$\sup_{\rho,\,v^*} \mathcal{R}_{\mathrm{det}} \approx 0.3479, \tag{S7}$$

approached as $\rho \to \infty$ at a threshold offset δ (the distance from the operating point to the inflection) with $\delta/2\varepsilon = 0.523$. There is no finite optimal transfer width: narrower is always better, and reaching 95% of the ceiling requires $u \gtrsim 6$, that is $1/\rho \lesssim \varepsilon/6$.

The limiting detector switches across its full range. At the optimum, and at the illustrative fast-access polarization $\varepsilon \simeq 0.1$ mV of §5, its open-loop beat drive exceeds the physiological-slope sigmoid's by some three orders of magnitude for the same e_0 and waveform, because the field excursion is about twice the threshold offset and the transfer switches across its full range instead of responding linearly. The disparity grows as the field weakens, the smooth sigmoid's response falling as ε^2 while a threshold repositioned with the field keeps its excursion.

The required transfer width is small: at $\varepsilon = 0.1$ mV the ceiling is approached only for $1/\rho \lesssim 17\ \mu$V; at $\varepsilon = 0.02$ mV the corresponding width is about 3 μV. Both follow from $1/\rho \lesssim \varepsilon/6$, with the threshold offset held to the same order as the applied polarization. A population sigmoid averages the spread of thresholds and noise; the JR voltage scale is $1/\rho \approx 1.8$ mV [21]. A microvolt width lies far below that and reads as an idealized microscopic threshold or rectifier rather than as a neural-mass sigmoid; real thresholds are themselves noisy and state-dependent, so it is not a measured single-cell property either. The regime in which the family approaches its ceiling is thus one in which the "population sigmoid" has ceased to describe a population.

The limit is a step detector, not an envelope follower. For a unit step at offset δ the carrier-averaged transduction is $\pi^{-1}\arccos(\delta/\mathcal{A})$ for $\mathcal{A} > \delta$ and zero below; with $\beta = \delta/2\varepsilon$ its tACS fundamental is $(2/\pi)\sqrt{1-\beta^2}$ and its TI coefficient reduces by parts to $\mathcal{R}_{\mathrm{det}} = 2\beta\,[\mathsf{K}(k_{\mathrm{ell}}^2) - \mathsf{E}(k_{\mathrm{ell}}^2)]/(\pi\sqrt{1-\beta^2})$

with elliptic parameter $k_{\mathrm{ell}}^2 = 1 - \beta^2$ and K, E the complete elliptic integrals, giving 0.3479 at $\beta = 0.523$ by direct quadrature and by the elliptic form alike (`threshold_bound_check.py`). At that optimum the step's second-to-first harmonic magnitude ratio is 0.151, not the 1/5 of a linear envelope follower. A bounded nondecreasing transfer is, up to a constant, a positive mixture of shifted steps. Both signed Fourier coefficients are linear in the transfer, and each step's direct-drive coefficient is nonnegative. The triangle inequality therefore bounds the mixture's amplitude ratio by the largest step ratio, whenever the denominator is nonzero. This extends the result beyond the logistic family; the maximum itself is a numerical evaluation, not a certified bound.

S5.1 Experimental amplitude and phase-locking comparisons

The protocols in Vieira et al. [22] do not match local polarization at the detector. The two-pair TI comparison gave a phase-locking effect ratio of 0.34 against tACS (95% confidence interval 0.04–0.65, 65 of 154 neurons), without correcting for TI currents up to ±2.5 mA per pair versus ±1 mA for tACS. A separate single-pair AM-tACS control compensated for frequency-dependent shunting to match extracellular modulation depth. Of twenty cells, ten responded to either modality and seven to both; in the latter group, median phase-locking value (PLV) was 0.10 for AM-tACS and 0.20 for tACS, against 0.043 at baseline. The reported TI modulation depths were about 0.5–0.62 V/m.

S6 Extension: the laminar column at low carriers

The two-band laminar column (LaNMM) adds a resonant intermediary between the field and the slow generator. These sweeps use 25–80 Hz carriers and polarizations up to 3 mV on P_2, exceeding the sigmoid scale $1/\rho \approx 1.8$ mV. They describe nonlinear frequency selection below TI's kHz range. The weak-field square law is tested separately in JR at 0.05–0.2 mV, one to two orders of magnitude below that scale (Fig. 8b).

S6.1 Model and measurements

The laminar extension (LaNMM) [23, 24] adds coupled alpha and gamma circuits: deep pyramidal P_1, spiny-stellate (SS), and somatostatin (SST) populations, and a superficial pyramidal P_2/PV circuit. We simulate this laminar neural mass model (LaNMM). We integrate the full 28-variable column [23, 24], with the field entering as in the single JR column, as a membrane polarization added inside the sigmoid argument of the target population. Its output rate φ_X (s^{-1}) is

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\}, \tag{S8}$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude ε in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S2). A perturbation of an external input *rate* would require clipping at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at Δf into the drive before the population nonlinearity received it. A signed membrane polarization carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, ε is directly comparable between the two models.

Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid simultaneously, as in the JR sweeps. The maps report the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials as the mean squared magnitude of the Hann-windowed fast Fourier transform (FFT) over the band, in arbitrary units, over a post-transient window. This measure includes autonomous and driven activity and does not isolate a phase-coherent response to stimulation.

Two sweeps probe the column: an amplitude–beat sweep over $(\varepsilon, \Delta f)$ at fixed carrier, and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed ε, built on the LaNMM module of [23]. The maps (Figs. S1, S7) show the *signed* stimulation-induced change, alpha power minus the unstimulated ($\varepsilon = 0$) baseline, on a zero-centered scale; over most of the P_1-driven grids the change is negative. For the alpha resonance we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (main-text Eq. (18)) of v_{P_1} at Δf.

The additional carrier comparison below uses the lock-in of v_{P_1}, averaged at each carrier over 17 beats: 7–9 and 12.25–14 Hz in 0.25 Hz steps. These lie more than 1.5 Hz from the field-free alpha peak near 10.6 Hz. We denote this arithmetic mean by $S(f_c)$. The carrier grid is 25–80 Hz in 1 Hz steps, with equal tone amplitudes $\varepsilon = 1.5$ mV for P_2 drive and 0.5 mV for P_1 drive. After 2 s of settling, we compare 3 and 10 s measurement windows. Field-free controls give mean background amplitudes of 0.0048 and 0.0012 mV over the same beats. This background includes a persistent component near 8 Hz, so it is not solely finite-window leakage. The statistic compares carrier profiles within each drive condition; the unequal amplitudes do not support a direct sensitivity comparison between targets.

S6.2 Frequency selection and carrier dependence

Driving either population with the two-tone field of Eq. (S8) and reading alpha-band power in the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials recovers the beat resonance (§S6.1, Fig. S2). Direct P_1 drive reproduces the JR alpha-band resonance on both stable-focus and limit-cycle sides; that sweep and its code are in the repository. The additional result is carrier selectivity through the coupled gamma circuit.

Table S5: LaNMM carrier profiles measured away from alpha. $S(f_c)$ is the mean lock-in amplitude of v_{P_1} over the 17 beats specified in §S6.1. Percentages are normalized to each row's own carrier maximum; the range and median use carriers from 30 to 80 Hz.

Drive	Window	Peak S (mV)	Peak f_c (Hz)	Range (%)	Median (%)
P_2, 1.5 mV	3 s	0.450	39	23–100	42
P_2, 1.5 mV	10 s	0.450	39	23–100	42
P_1, 0.5 mV	3 s	0.092	25	42–68	53
P_1, 0.5 mV	10 s	0.091	25	42–67	52

Sweeping the field amplitude ε and beat frequency Δf at a fixed carrier produces a region of enhanced alpha-band power: the response in P_1 is largest when Δf matches the deep alpha ($\sim$10 Hz) and grows with ε (Figs. S1a,b and S7a,b). The dominant-frequency contours identify where the output is close to the beat, without establishing phase locking of the autonomous rhythm. Away from the enhancement region the drive *reduces* alpha-band power: with the field on P_1, 95% of the amplitude–beat grid and 97% of the carrier–beat grid lie below the field-free baseline (Fig. S7), the drive pulling the autonomous alpha out of the 8–12 Hz band or suppressing it; with the field on P_2 the fractions are 44% and 31% (Fig. S1).

The carrier dependence illustrates the two coupling limits of §2.2–§2.8 within this model's 25–80 Hz carrier range. When the field is delivered to the superficial loop (P_2), alpha-band power enhancement in P_1 is largest in the gamma carrier band with Δf near the P_1 alpha (Fig. S1c,d). At $\Delta f = 10$ Hz, the carrier maximum is 44 Hz. This selectivity is consistent with a double resonance: the fast loop acts as a tuned radio-frequency (RF) front end whose dynamics affect the carrier dependence of the deep response. With direct P_1 drive, a ridge near the column's own alpha persists across the carrier range (Fig. S7c). That ridge contains autonomous activity and is not a measurement of the demodulated component alone.

The lock-in also contains the autonomous rhythm when the beat is close to alpha; its maximum over beats cannot isolate the driven response. We therefore compare the off-alpha statistic $S(f_c)$ defined in §S6.1 (Table S5). With P_2 drive it peaks near 39 Hz and falls to about 16–17% of that peak at a 25 Hz carrier. With P_1 drive it peaks at 25 Hz and spans about 42–68% of that peak over 30–80 Hz. These are properties of the specified amplitudes and beat average, not universal carrier-rejection factors. Single-beat profiles have deeper minima, including a dip near a 45 Hz carrier under direct P_1 drive: the gamma circuit can affect the response in either stimulation condition.

The carrier peaks and aggregate ranges persist with the longer window, although individual normalized values change by up to 3.1 percentage points for P_2 drive and 9.6 for P_1 drive. The smallest S exceeds its mean field-free background by factors of about 16 and 8, respectively, with the 3 s window, and 62 and 32 with 10 s. Some individual beat–carrier points remain near or below background; the comparison concerns the mean over beats, not each grid point. TI's kHz carriers lie outside this model's range. Within it, the carrier can excite a network mode and affect the detected response, qualifying the inert-carrier limit of §2.7 and linking it to frequency-tagging and intermodulation paradigms [25].

Within the 25–80 Hz carrier range and at the finite polarizations used here, the LaNMM shows alpha-band power changes concentrated near the intrinsic alpha, amplitude-dependent widening of the frequency-dominance region, and enhanced responses when superficial drive passes through the gamma circuit. The power maps include autonomous and driven activity; they do not isolate TI efficacy. The gamma-carrier preference, around 40 Hz rather than kHz, remains a separate prediction to test. Esmaeilpour et al. [6] swept 0.1–2 kHz carriers at a fixed 5 Hz beat against carbachol-induced gamma, finding decreasing sensitivity with carrier frequency. Those carriers did not sample the gamma band. Violante et al. [13] also used a fixed 5 Hz beat. Neither study tests

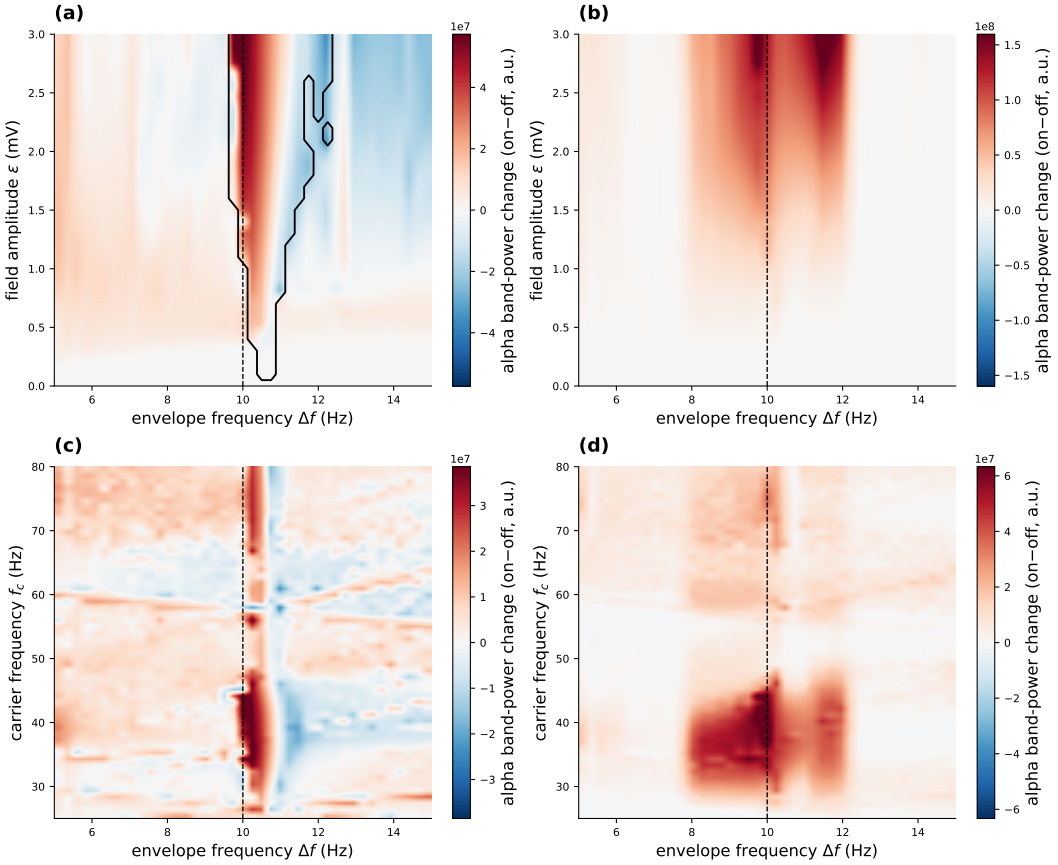


Figure S1: LaNMM frequency selection: two-tone TI field on the superficial P_2 (gamma) loop. The field enters the P_2 sigmoid argument as a membrane polarization ε (mV) with all external drives flat. Color shows the signed change of alpha-band (8–12 Hz) power from the unstimulated $\varepsilon = 0$ baseline, on a zero-centered scale (arbitrary units). *(a,b)* P_1 and P_2 enhancement vs. ε and Δf (carrier 40 Hz), peaking at the P_1 alpha (10.0 Hz, dashed). The black contour in *(a)* is the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz, drawn because band power alone cannot separate a driven from an autonomous rhythm; it is a dominance screen, not a phase-capture test. The $\varepsilon = 0$ row is excluded, since with no drive the autonomous 10.67 Hz alpha already lies within the tolerance of two grid points (a resolution floor of 0.5 Hz); the set widens to 2.75 Hz at 3 mV. *(c,d)* Enhancement vs. carrier and Δf at $\varepsilon = 1.5$ mV: P_1 alpha-band enhancement is largest in the gamma carrier band with Δf near the P_1 alpha. At $\Delta f = 10$ Hz, the carrier maximum is 44 Hz. The field-free P_2 gamma peak is near 39.7 Hz; the finite-drive maximum need not coincide with it.

this low-carrier resonance prediction.

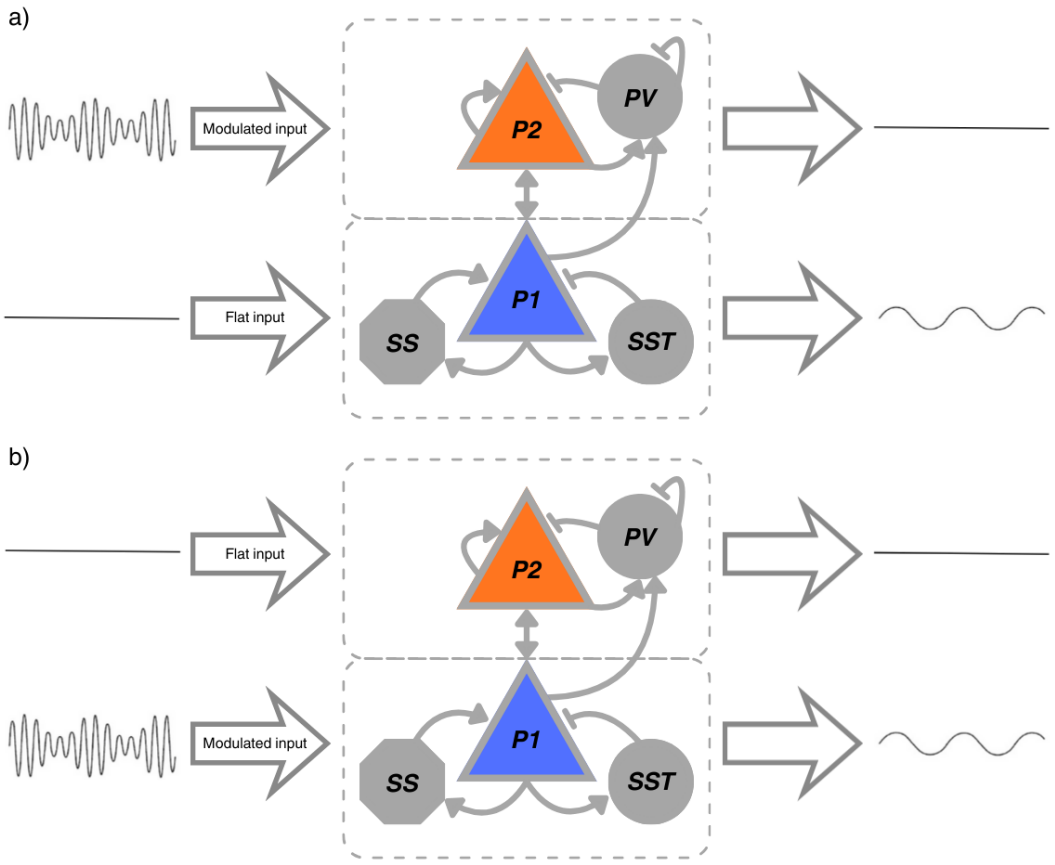


Figure S2: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the TI field drives P_2 while P_1 receives a flat drive; *(b)* the TI field drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the generated beat-frequency component is transferred to the deep loop: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency component (right traces).

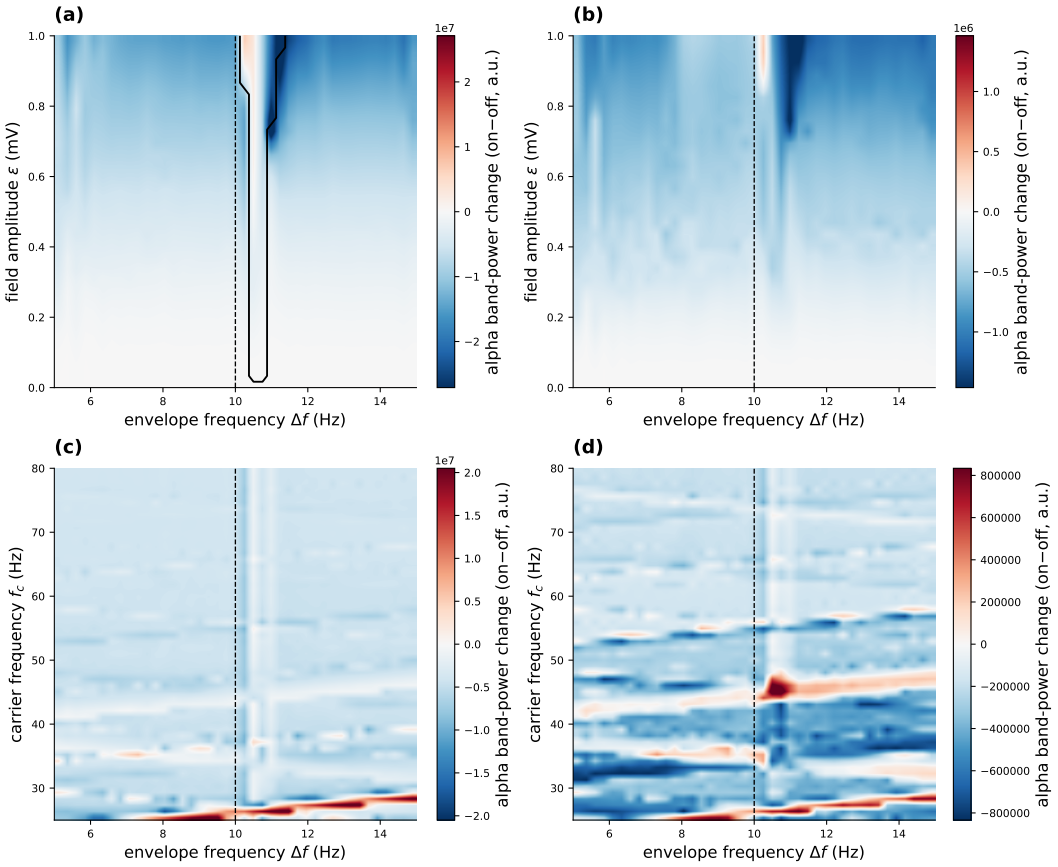


Figure S7: LaNMM frequency selection: two-tone TI field delivered directly to the deep P_1 (alpha) loop. As Fig. S1 but with the field on P_1. The deep loop shows frequency dominance at smaller fields, so the amplitude range is $\varepsilon \leq 1$ mV here against 3 mV there. These finite-amplitude maps extend beyond the perturbative regime of the theory. Color: signed change of alpha-band power from the $\varepsilon = 0$ baseline, zero-centered; most of the grid is suppression. *(a,b)* P_1 and P_2 enhancement vs. ε and Δf; the black contour in *(a)* outlines the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (§S7), computed from the dominant output frequency because band power alone cannot separate a driven from an autonomous rhythm; the $\varepsilon = 0$ row is excluded (resolution floor 0.5 Hz). The set widens to 1.25 Hz at $\varepsilon = 1$ mV. *(c,d)* Enhancement vs. carrier frequency and Δf at $\varepsilon = 0.5$ mV. The ridge near alpha persists across the carrier band and includes the autonomous rhythm. A separate lock-in comparison away from alpha, including its window and background controls, is given in Table S5; these panels show band-power changes.

S7 Experimental tests and numerical definitions

Forced response. Below the Hopf the column does not oscillate on its own; we quantify its response by the lock-in amplitude at the beat Δf (main-text Eq. (18)). There is no autonomous rhythm to entrain in the stable-focus J-curve, timing, and tACS sweeps.

Frequency dominance and entrainment. Above the Hopf the column free-runs at f_0. A 1:1 phase-capture test would follow the unwrapped phase difference from the drive and require absence of secular drift. Our maps instead report $|f_{\text{out}} - \Delta f| < 0.3$ Hz, where f_{out} is the largest non-DC FFT peak. This is a frequency-dominance screen: a forced component can outgrow a persisting autonomous one without capturing it. When the components coexist, instantaneous-phase estimates also require their separation [26]. The maps therefore identify a dominance region, not a measured Arnold tongue.

Beat-locked rate and gating. Gating means that the amplitude of a persisting rhythm is modulated by the beat without frequency capture [27, 28]. The QIF statistic used here is the lock-in of the population rate r_E at Δf, complemented by the rate folded on beat phase over whole cycles. It measures beat-locked firing. A raw rate fold does not extract the gamma amplitude, and sidebands can arise from amplitude or phase modulation. The above-Hopf QIF panels therefore establish coexistence and mixing of the beat and gamma rhythms; a gamma-amplitude estimate would be needed to identify gating specifically.

The protocol for Fig. S4 places the column on the limit-cycle side (input p on the stable alpha-cycle branch below its upper Hopf), drives it with $s(t) = \varepsilon \cos \Omega t$, and records the dominant frequency. Grid widths and the clipping check are specified in the code. As the Hopf is approached, the autonomous cycle shrinks and the frequency-dominance region broadens in these simulations. Its width and the stable-focus susceptibility describe different responses; their increase near onset does not make them the same quantity.

S7.1 Lock-in leakage control

Out-of-band components can leak into the estimate, particularly the DC level of the JR voltage readout ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). The finite-window projection of a constant onto the quadrature references need not vanish, so we subtract $\bar{v}$ explicitly; this makes the metric accurate down to the 10^{-8} level needed for the JR kHz analysis.

S7.2 JR integration and measurement windows

We integrate with a fixed-step fourth-order Runge–Kutta (RK4) scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S9 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and 1.5×10^{-5} s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.8 are analytic evaluations of Eq. (13), not simulations.

The resonance sweep uses 10 s of settling and a 20 s measurement window. Long windows reduce transients from weakly damped modes and average beating when autonomous activity persists. Other runs use the windows specified in their scripts. The two-tone 2 kHz example (Fig. S6) uses $p = 400$ s^{-1}, $C = 135$, and 0.5 mV per tone with $\tau = 0.2$ ms. From a zero initial state, it settles for 4 s and records 4 s at $\Delta t = 1.5 \times 10^{-5}$ s, sampling before each RK4 step. Its amplitude spectra use plain-mean subtraction and a Hann window. Parameter sweeps advance all N_{grid} combinations simultaneously in a vectorized $(N_{\text{grid}}, 6)$ state array.

S7.3 Evidence and measurement design

The detailed matrix supports the compact experimental tests in article Table 4. Model comparisons and biological observations are distinguished; the latter do not calibrate the detector.

Table S6: Predictions, existing evidence, and tests. Published observations provide qualitative comparisons; their quantitative reanalysis remains to be done.

Prediction (model basis)	Existing evidence	Proposed test
Carrier dependence. At fixed local polarization, the leading beat is carrier-independent in the static limit and fast-carrier QIF asymptote, provided no other circuit mode is excited. Access restores dependence at fixed applied field; detector dynamics can add it near the crossover.	Network sensitivity, cellular mixing, and thresholds depend on carrier frequency [6, 29, 30, 31]; bare kHz fields have limited effects on hippocampal oscillations [32]. Local polarizations were not held fixed.	Measure each local polarization while sweeping f_c at fixed beat and state; normalize the weak beat by their product. Hold network gain fixed or measure it independently to separate access from residual detector dependence. Measure any residual cutoff rather than assigning it to a compartment.
Independent carrier amplitudes. At fixed detector and network gains, quadratic modulation excursion follows $2\lvert\varepsilon_1\varepsilon_2\rvert$; an ideal envelope follower tracks $2\min(\lvert\varepsilon_1\rvert, \lvert\varepsilon_2\rvert)$ (Eq. (9), Table 1).	The unequal-amplitude spatial example separates these maps; a biological test with independent local-polarization calibration is not supplied here.	Hold the weaker polarization fixed and vary the stronger, then swap tones. Keep both inputs weak, frequencies fixed, and the operating state controlled. Reconstruct the slow-response excursion after correcting network filtering; a beat lock-in alone does not measure the envelope's full excursion.
Current-ratio steering. With fixed electrode pairs, frequencies, access, and neural state, the quadratic beat map scales with $I_1^{\mathrm{pair}} I_2^{\mathrm{pair}}$; its normalized pattern does not move. An envelope follower can instead shift its maximum.	The product-law restriction [33] holds in IXI002 (§4.4); a physiological beat-map test remains open.	Vary the current ratio at fixed electrodes and compare normalized beat maps and target-to-reference ratios, using a cortex-only reference to test deep-target selectivity. Keep the projection direction fixed at each site, including any direction chosen to maximize the envelope at a reference ratio. Measure the DC shift and repeat at lower amplitude to check state dependence.
State and carrier dependence. The static sigmoid follows σ'' and reverses sign at its inflection (PEIX$= 0$). Fast-carrier QIF mixes in voltage and reads the source through rate and network susceptibility; its DC term follows F' without the same sign rule (§2.10).	Cell-type-specific responses (PV versus pyramidal) are reported [34]; the detector coefficient remains model-dependent.	Vary tonic drive and carrier frequency at fixed beat; measure beat amplitude and phase. Compare with the predicted coefficient after accounting for network susceptibility. Amplitude changes alone do not isolate detection.

Continued on next page

Table S6 continued

Prediction (model basis)	Existing evidence	Proposed test
Transduction class. At a matched site, weak smooth detection with nonzero curvature gives $Y_\Omega \propto \varepsilon^2$ and a harmonic ratio vanishing as $\varepsilon \to 0$. An ideal envelope follower gives $Y_\Omega \propto \varepsilon$ and ratio 1/5; a threshold has onset $2\varepsilon = \delta$ and a variable ratio.	Untested with local-polarization calibration; existing comparisons match extracellular field rather than ε.	Sweep amplitude off resonance, reading both the log–log slope and $\lvert Y_{2\Omega}/Y_\Omega \rvert$. Near resonance, suppression by approximately $3\mathcal{Q}$ requires correction with an independently measured $G(2\Omega)/G(\Omega)$ (§3.2).
Frequency selectivity. The beat response peaks at the region's resonant rhythm and grows as $1/\gamma$ near its Hopf; state changes can alter both the generated drive and the recurrent gain [23].	The quantitative gain–bandwidth relation is not tested here in biological data.	Sweep Δf through the intrinsic band across brain states and track the peak frequency, height, and width.
Population modulation and mean rate. Beat modulation can exceed the mean shift; their ratio peaks and then falls as the amplified oscillation rectifies (§4.6–4.9).	Qualitatively consistent with TI altering spike timing more than mean rate in primates [22].	Compare beat lock-in with mean-rate change in commensurate effect measures, retaining the finite-amplitude qualification.
Network readout. Population nonlinearity and synaptic recurrence contribute to the measured response.	Most pyramidal TI responses vanish under synaptic blockade [34], supporting network involvement without excluding cellular mixing.	Test whether synaptic blockade reduces recurrent amplification while preserving a cellular detector.
Equal-RMS carrier control. At second order, equal local RMS polarization gives the same steady source when the carriers see the same DC coefficient. TI also supplies a beat; either waveform can induce an oscillation if the mean shift crosses a bifurcation.	Demonstrated in the PING mean field (Fig. 14c,d). The corresponding spiking-network and in vivo controls are not provided here.	Compare two tones with one carrier of matched RMS polarization and duration, checking the DC coefficient across the carrier interval. Measure mean activity, intrinsic oscillations, and beat lock-in separately to distinguish the steady shift from the beat response.

References

[1] Giulio Ruffini. Analysis and extension of exact mean-field theory with dynamic synaptic currents. *bioRxiv*, 2021. Preprint 2021.09.01.458563 (NMM2).

[2] Yair Deitcher, Guy Eyal, Lida Kanari, Matthijs B. Verhoog, Guy Antoine Atenekeng Kahou, Huibert D. Mansvelder, Christiaan P. J. de Kock, and Idan Segev. Comprehensive morpho-electrotonic analysis shows 2 distinct classes of L2 and L3 pyramidal neurons in human temporal cortex. *Cerebral Cortex*, 27(11):5398–5414, 2017.

[3] Guy Eyal, Matthijs B. Verhoog, Guilherme Testa-Silva, Yair Deitcher, Johannes C. Lodder, Ruth Benavides-Piccione, Juan Morales, Javier DeFelipe, Christiaan P. J. de Kock, Huibert D. Mansvelder, and Idan Segev. Unique membrane properties and enhanced signal processing in human neocortical neurons. *eLife*, 5:e16553, 2016.

[4] Charles C. H. Cohen, Marko A. Popovic, Jan Klooster, Marie-Theres Weil, Wiebke Möbius, Klaus-Armin Nave, and Maarten H. P. Kole. Saltatory conduction along myelinated axons involves a periaxonal nanocircuit. *Cell*, 180(2):311–322.e15, 2020.

[5] Maarten H. P. Kole, Susanne U. Ilschner, Björn M. Kampa, Stephen R. Williams, Peter C. Ruben, and Greg J. Stuart. Action potential generation requires a high sodium channel density in the axon initial segment. *Nature Neuroscience*, 11(2):178–186, 2008.

[6] Zeinab Esmaeilpour, Greg Kronberg, Davide Reato, Lucas C. Parra, and Marom Bikson. Temporal interference stimulation targets deep brain regions by modulating neural oscillations. *Brain Stimulation*, 14(1):55–65, 2021.

[7] Charlotte E. Luff, Patrycja Dzialecka, Emma Acerbo, Adam Williamson, and Nir Grossman. Pulse-width modulated temporal interference (pwm-ti) brain stimulation. *Brain Stimulation*, 17(1):92–103, January 2024.

[8] Iurii Semenov, Vitalii Kim, Giedre Silkuniene, and Andrei G. Pakhomov. Excitation and polarization of isolated neurons by high-frequency sine waves for temporal interference stimulation. *Cell Reports Physical Science*, 6(7):102660, 2025.

[9] Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.

[10] Jesper D. Nielsen, Kristoffer H. Madsen, Oula Puonti, Hartwig R. Siebner, Christian Bauer, Camilla Gøbel Madsen, Guilherme B. Saturnino, and Axel Thielscher. Automatic skull segmentation from MR images for realistic volume conductor models of the head: assessment of the state-of-the-art. *NeuroImage*, 174:587–598, 2018.

[11] Christian Gaser, Robert Dahnke, Paul M. Thompson, Florian Kurth, Eileen Luders, and Alzheimer's Disease Neuroimaging Initiative. CAT: a computational anatomy toolbox for the analysis of structural MRI data. *GigaScience*, 13:giae049, 2024.

[12] Ricardo Salvador, Junhong Zhou, Brad Manor, and Giulio Ruffini. Group optimization methods for dose planning in tES. *Journal of Neural Engineering*, 22:046045, 2025.

[13] Ines R. Violante, Ketevan Alania, Antonino M. Cassarà, Esra Neufeld, Emma Acerbo, Romain Carron, Adam Williamson, Danielle L. Kurtin, Edward Rhodes, Adam Hampshire, Niels Kuster, Edward S. Boyden, Alvaro Pascual-Leone, and Nir Grossman. Non-invasive temporal interference electrical stimulation of the human hippocampus. *Nature Neuroscience*, 26(11):1994–2004, 2023.

[14] K. Yatsuda, W. Yu, and J. Gomez-Tames. Population-level insights into temporal interference for focused deep brain neuromodulation. *Frontiers in Human Neuroscience*, 18:1308549, 2024.

[15] O. C. Zienkiewicz and J. Z. Zhu. The superconvergent patch recovery and a posteriori error estimates. Part 1: The recovery technique. *International Journal for Numerical Methods in Engineering*, 33(7):1331–1364, 1992.

[16] Thomas Radman, Raddy L. Ramos, Joshua C. Brumberg, and Marom Bikson. Role of cortical cell type and morphology in subthreshold and suprathreshold uniform electric field stimulation in vitro. *Brain Stimulation*, 2(4):215–228.e3, October 2009.

[17] Marom Bikson, Masashi Inoue, Hiroki Akiyama, Jackie K. Deans, John E. Fox, Hiroyoshi Miyakawa, and John G. R. Jefferys. Effects of uniform extracellular dc electric fields on excitability in rat hippocampal slices in vitro. *The Journal of Physiology*, 557(1):175–190, May 2004.

[18] Jacqueline K. Deans, Andrew D. Powell, and John G. R. Jefferys. Sensitivity of coherent oscillations in rat hippocampus to ac electric fields. *The Journal of Physiology*, 583(2):555–565, August 2007.

[19] Nir Grossman, David Bono, Nina Dedic, Suhasa B. Kodandaramaiah, Andrii Rudenko, Ho-Jun Suk, Antonino M. Cassara, Esra Neufeld, Niels Kuster, Li-Huei Tsai, Alvaro Pascual-Leone, and Edward S. Boyden. Noninvasive deep brain stimulation via temporally interfering electric fields. *Cell*, 169(6):1029–1041.e16, 2017.

[20] Bruce Fischl, David H. Salat, Evelina Busa, Marilyn Albert, Megan Dieterich, Christian Haselgrove, Andre van der Kouwe, Ron Killiany, David Kennedy, Shuna Klaveness, Albert Montillo, Nikos Makris, Bruce Rosen, and Anders M. Dale. Whole brain segmentation: automated labeling of neuroanatomical structures in the human brain. *Neuron*, 33(3):341–355, 2002.

[21] Ben H. Jansen and Vincent G. Rit. Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366, 1995.

[22] Pedro G. Vieira, Matthew R. Krause, and Christopher C. Pack. Temporal interference stimulation disrupts spike timing in the primate brain. *Nature Communications*, 15:4558, 2024.

[23] Giulio Ruffini, Edmundo Lopez-Sola, Raul Palma, Roser Sanchez-Todo, Jakub Vohryzek, Francesca Castaldo, and Karl Friston. Cross-frequency coupling as a neural substrate for prediction error evaluation: a laminar neural mass modeling approach. *Neural Computation*, 38(8):1299–1343, 2026.

[24] Roser Sanchez-Todo, André M. Bastos, Edmundo Lopez-Sola, Borja Mercadal, Emiliano Santarnecchi, Earl K. Miller, Gustavo Deco, and Giulio Ruffini. A physical neural mass model framework for the analysis of oscillatory generators from laminar electrophysiological recordings. *NeuroImage*, 270:119938, 2023.

[25] Nathan Gordon, Jakob Hohwy, Matthew J. Davidson, Jeroen J. A. van Boxtel, and Naotsugu Tsuchiya. From intermodulation components to visual perception and cognition—a review. *NeuroImage*, 199:480–494, 2019.

[26] Jean-Philippe Lachaux, Eugenio Rodriguez, Jacques Martinerie, and Francisco J. Varela. Measuring phase synchrony in brain signals. *Human Brain Mapping*, 8(4):194–208, 1999.

[27] Ryan T. Canolty and Robert T. Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010.

[28] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010.

[29] Charlotte E. Luff, Robert Peach, Emma-Jane Mallas, Edward Rhodes, Felix Laumann, Edward S. Boyden, David J. Sharp, Mauricio Barahona, and Nir Grossman. The neuron mixer and its impact on human brain dynamics. *Cell Reports*, 43(6):114274, 2024.

[30] Aleksandar Opančar, Petra Ondráčková, David Samuel Rose, Jan Trajlinek, Vedran Đerek, and Eric Daniel Głowacki. The same biophysical mechanism is involved in both temporal interference and direct kHz stimulation of peripheral nerves. *Nature Communications*, 16:9006, 2025.

[31] Ryan B. Budde, Michael T. Williams, and Pedro P. Irazoqui. Temporal interference current stimulation in peripheral nerves is not driven by envelope extraction. *Journal of Neural Engineering*, 20(2):026041, 2023.

[32] Zeinab Esmaeilpour, Mark Jackson, Greg Kronberg, Tianhe Zhang, Rosana Esteller, Brad Hershey, and Marom Bikson. Limited sensitivity of hippocampal synaptic function or network oscillations to unmodulated kilohertz electric fields. *eNeuro*, 7(6):ENEURO.0368–20.2020, 2020.

[33] Antonino M. Cassarà, Taylor H. Newton, Katie Zhuang, Sabine J. Regel, Peter Achermann, Alvaro Pascual-Leone, Niels Kuster, and Esra Neufeld. Recommendations for the safe application of temporal interference stimulation in the human brain part II: Biophysics, dosimetry, and safety recommendations. *Bioelectromagnetics*, 46(1):e22536, 2025.

[34] Sara Caldas-Martinez, Chaitanya Goswami, Mats Forssell, Jiaming Cao, Alison L. Barth, and Pulkit Grover. Cell-specific effects of temporal interference stimulation on cortical function. *Communications Biology*, 7:1076, 2024.